Vector & Three-Dimensional Geometry

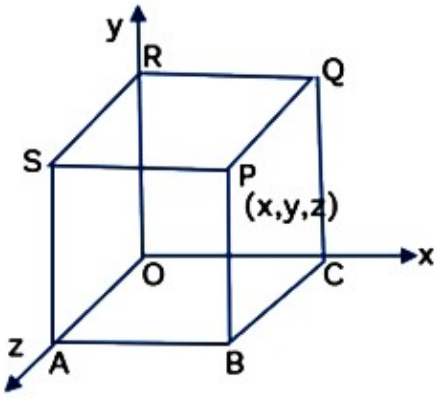
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Three Dimensional Geometry

VECTORS AND SCALARS :

COORDINATES OF A POINT IN SPACE:



Consider a point P in space whose position is given by (x,y,z) where x,y,z are perpendicular distance from yz plane, zx plane and xy plane respectively.

If we assume $\hat{i},\hat{j},\hat{k}$ unit vectors along OX,OY,OZ respectively then the position vector of point P is $x\hat{i}+y\hat{j}+z\hat{k}$ or simply (x,y,z).

When a point lies on Co-ordinates

(I) $x-$axis $(\alpha,0,0)$

(ii) $y-$axis $(0,\beta,0)$

(iii) $z-$axis $(0,0,\gamma)$

(IV) XY–plane $(\alpha,\beta,0)$

(V) XZ-plane $(\alpha,0,\gamma)$

(VI) YZ–plane $(0,\beta,\gamma)$

Distance formulae:

Distance between the points (x_1,y_1,z_1) and (x_2,y_2,z_2) $=\sqrt{(x_1-x_2)^2+(y_1-y_2)^2+(z_1-z_2)^2}$

Section formulae:

(1) Coordinates of a point P which divides line joining $A(x_1,y_1,z_1)$ and $B(x_2,y_2,z_2)$ in the ratio m : n internally is given by $\left(\frac{mx_2+nx_1}{m+n},\frac{my_2+ny_1}{m+n},\frac{mz_2+nz_1}{m+n}\right)$

(2) Coordinates of a point P which divides line joining $A(x_1,y_1,z_1)$ and $B(x_2,y_2,z_2)$ in the ratio m : n externally is given by $\left(\frac{mx_2-nx_1}{m-n},\frac{my_2-ny_1}{m-n},\frac{mz_2-nz_1}{m-n}\right)$.

Direction cosines:

If α, β, γ are the angles which vector $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$ makes with positive direction of the x, y, z axes respectively then α, β, γ are called direction angles and their cosines $\cos\alpha, \cos\beta, \cos\gamma$ are called the direction cosines of the vector and are generally denoted l, m, n respectively.

Thus $l = \cos\alpha, m = \cos\beta, n = \cos\gamma$.

Point to Remember!!!

Coordinates of mid-point of line joining $A(x_1, y_1, z_1)$ and $B(x_2, y_2, z_2)$ is $\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}, \frac{z_1 + z_2}{2}\right)$

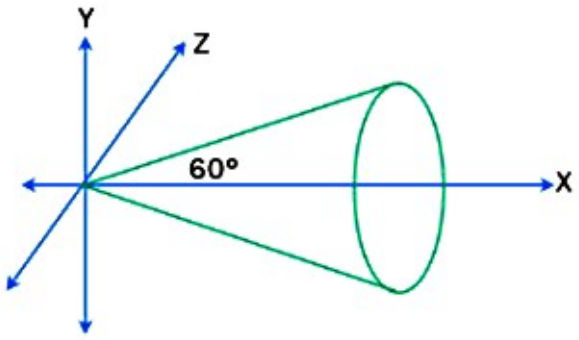
Note :

(i) If a line makes angle α, β, γ with positive direction of x, y, z axes respectively then direction cosines of line will be $\cos\alpha, \cos\beta, \cos\gamma$ or $-\cos\alpha, -\cos\beta, -\cos\gamma$

(ii) A unit vector along the line whose direction cosines are $\cos\alpha, \cos\beta, \cos\gamma$ can be written as $(\cos\alpha)\hat{i} + (\cos\beta)\hat{j} + (\cos\gamma)\hat{k}$.

(iii) If a vector $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$ makes angles α, β, γ with positive direction of x, y, z axes respectively then

$\cos\alpha = \frac{\vec{a} \cdot \hat{i}}{|\vec{a}||\hat{i}|} = \frac{a_1}{|\vec{a}|}, \cos\beta = \frac{\vec{a} \cdot \hat{j}}{|\vec{a}||\hat{j}|} = \frac{a_2}{|\vec{a}|}$ and $\cos\gamma = \frac{\vec{a} \cdot \hat{k}}{|\vec{a}||\hat{j}|} = \frac{a_3}{|\vec{a}|}$



$\therefore \cos^2\alpha + \cos^2\beta + \cos^2\gamma = \frac{a_1^2 + a_2^2 + a_3^2}{|\vec{a}|^2} \Rightarrow \cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$

Also note that $\sin^2\alpha + \sin^2\beta + \sin^2\gamma = 2$

(iv) Direction cosines of x -axis are $(1,0,0)$ or $(-1,0,0)$.

Direction cosines of y -axis are $(0,1,0)$ or $(0,-1,0)$.

Direction cosines of z-axis are $(0,0,1)$ or $(0,0,-1)$.

Direction ratios :

If a, b, c are three numbers proportional to the direction cosines l, m, n of a straight line, then a, b, c are called its direction ratios. They are also called direction numbers or direction components.

Hence, we have $\frac{l}{a} = \frac{m}{b} = \frac{n}{c} = \lambda$(say) $\Rightarrow l = a\lambda, m = b\lambda, n = c\lambda$

$\because l^2 + m^2 + n^2 = 1 \Rightarrow \left(a^2 + b^2 + c^2\right)\lambda^2 = 1 \Rightarrow \lambda = \pm\frac{1}{\sqrt{a^2 + b^2 + c^2}}$

$\therefore l = \pm\frac{a}{\sqrt{a^2 + b^2 + c^2}}, m = \pm\frac{b}{\sqrt{a^2 + b^2 + c^2}}$ and $n = \pm\frac{c}{\sqrt{a^2 + b^2 + c^2}}$

Note :

(i) Direction ratios of a line is not unique but infinite in number but direction cosines will be for a line will be only two. (l, m, n or –l,–m,–n)

(ii) Direction ratios of a line joining two points A and B are proportional to $x_2 - x_1, y_2 - y_1, z_2 - z_1$.

(iii) Projection of a Point on a Line:
Let P be a point and AB be a given line. Draw perpendicular PQ from P on AB which meets it at Q. This point Q is called projection of P on the line AB.

(iv) Projection of a Line Segment Joining Two Points on a Line:
Projection of the line segment joining two points $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$ on another line whose direction cosines are l, m, n is $AB = \left|l(x_2 - x_1) + m(y_2 - y_1) + n(z_2 - z_1)\right|$.

Point to Remember!!!

A vector along the line with direction ratios a, b, c can be $a\hat{i} + b\hat{j} + c\hat{k}$.

Proof:

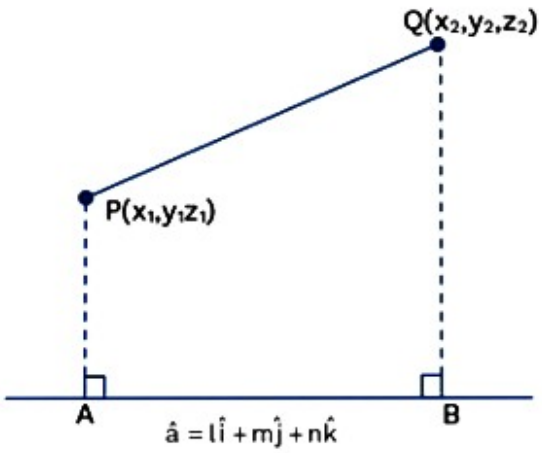
Vector $\overline{PQ} = (x_2 - x_1)\hat{i} + (y_2 - y_1)\hat{j} + (z_2 - z_1)\hat{k}$

A unit vector along another line $\hat{a} = l\hat{i} + m\hat{j} + n\hat{k}$

$\therefore$ Projection AB = Projection of

$\overline{PQ}$ on $\hat{a} = \frac{\left|\overline{PQ} \cdot \hat{a}\right|}{\left|\hat{a}\right|}$

$= \left|l(x_2 - x_1) + m(y_2 - y_1) + n(z_2 - z_1)\right|$



Q. **Find the D.C. of the vector $2\hat{i} - 2\hat{j} - \hat{k}$**

Sol. Let $\vec{a} = 2\hat{i} - 2\hat{j} - \hat{k}$

$|\vec{a}| = \sqrt{(2)^2 + (-2)^2 + (-1)^2} = 3$

D.C. of $\vec{a}$ are $\frac{2}{3}, \frac{-2}{3}, \frac{-1}{3}$ (True)

Q. **Find the locus of all points P for which $\overline{OP}$ represents a vector with direction cosine $\cos\alpha = \frac{1}{2}$ (O is origin).**

Sol. From the diagram, it is obvious that locus is a cone concentric with the positive x-axis having vertex at the origin and the slant height equal to the magnitude of the vector.

Q. **Find the number of unit vectors with $\cos\alpha = \frac{1}{2}$ and $\cos\beta = \frac{1}{2}$.**

Sol. $\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$

or $\frac{1}{4} + \frac{1}{4} + \cos^2\gamma = 1$

or $\cos^2\gamma = 1 - \frac{1}{2} = \frac{1}{2}$

$\therefore \cos\gamma = \pm\frac{1}{\sqrt{2}}$

Two value of cos γ are possible. Hence, two vectors are possible, one corresponding to direction cosines $\frac{1}{2}, \frac{1}{2}, \frac{1}{\sqrt{2}}$ and other corresponding to direction cosines $\frac{1}{2}, \frac{1}{2}, -\frac{1}{\sqrt{2}}$.

Q. **Find the number of vector with direction angles $\alpha = 30°$ and $\beta = 30°$.**

Sol. $\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$

$\cos^2 30^\circ + \cos^2 30^\circ + \cos^2\gamma = 1$

$\left(\frac{\sqrt{3}}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2 + \cos^2\gamma = 1$

$$\cos^2\gamma = 1-2\left(\frac{3}{4}\right) = 1-\frac{3}{2} = \frac{-1}{2}$$

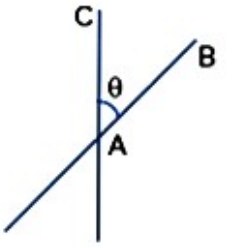
Not possible

$\therefore$ Required number of vectors is zero.

Angle between two lines :

If direction ratios of two lines are a_1, b_1, c_1 and a_2, b_2, c_2 then acute angle between two lines is given by

$$\cos\theta = \frac{\left|a_1a_2 + b_1b_2 + c_1c_2\right|}{\sqrt{a_1^2 + b_1^2 + c_1^2}\sqrt{a_2^2 + b_2^2 + c_2^2}}$$



Proof:

Vector along lines can be taken as $\vec{a} = a_1\hat{i} + b_1\hat{j} + c_1\hat{k}$ and $\vec{b} = a_2\hat{i} + b_2\hat{j} + c_2\hat{k}$.

Acute angle between lines = acute angle between vectors $\vec{a}$ and $\vec{b}$.

$$\therefore \cos\theta = \frac{\left|\vec{a}\cdot\vec{b}\right|}{\left|\vec{a}\right|\left|\vec{b}\right|} = \frac{a_1a_2 + b_1\ b_2 + c_1c_2}{\sqrt{a_1^2 + b_1^2 + c_1^2}\sqrt{a_2^2 + b_2^2 + c_2^2}}$$

If direction cosines of lines are $l_1,\ m_1, n_1$ and $l_2,\ m_2, n_2$ then acute angle between them is given by $\cos\theta = \left|l_1l_2 + m_1m_2 + n_1n_2\right|$

Point to Remember!!!

(i) If lines are perpendiculars (i.e. vectors along them are also perpendicular) then $a_1a_2 + b_1b_2 + c_1c_2 = 0$ or

$l_1l_2 + m_1m_2 + n_1n_2 = 0$

(ii) If lines are parallel (i.e. vectors along them are also parallel) then $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$

or $\frac{l_1}{l_2} = \frac{m_1}{m_2} = \frac{n_1}{n_2}$.

Q. **Find the coordinates of the point which divides the line joining points $(2,3,4)$ and $(3,-4,7)$ in ratio $5:3$ internally.**

Sol. Let the coordinates of the required point be (x,y,z) then $x = \frac{2(3)+3(5)}{3+5} = \frac{21}{8}$;

$y = \frac{3(3)-4(5)}{3+5} = \frac{-11}{8}; z = \frac{4(3)+7(5)}{3+5} = \frac{47}{8}$

$\therefore$ The required point is $\left(\frac{21}{8}, \frac{-11}{8}, \frac{47}{8}\right)$.

Q. **Find unit vector(s) with $\cos\alpha = \frac{1}{2}$ and $\cos\beta = \frac{1}{2}$, where α, β are angles made by unit vector with positive direction x, y axes respectively.**

Sol. Let unit vectors makes angle γ with positive z-axis.

$\therefore$ Unit vector will be $(\cos\alpha)\hat{i} + (\cos\beta)\hat{j} + (\cos\gamma)\hat{k}$

$\because \cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1 \Rightarrow \frac{1}{4} + \frac{1}{4} + \cos^2\gamma = 1$

$\Rightarrow \cos^2\gamma = \frac{1}{2} \Rightarrow \cos\gamma = \pm\frac{1}{\sqrt{2}}$

$\therefore$ Unit vector will be: $\frac{\hat{i}}{2} + \frac{\hat{j}}{2} \pm \frac{\hat{k}}{\sqrt{2}}$

Q. **Find the direction cosines of two lines which are connected by the relations $l - 5m + 3n = 0$ and $7l^2 + 5m^2 - 3n^2 = 0$**

Sol. The given relations are $l - 5m + 3n = 0 \Rightarrow l = 5m - 3n$ ...(1)

and $7l^2 + 5m^2 - 3n^2 = 0$ .....(2)

Putting the value of l from (1) in (2), we get

$7(5m - 3n)^2 + 5m^2 - 3n^2 = 0$

or $(2m - n)(3m - 2n) = 0 \Rightarrow \frac{m}{n} = \frac{1}{2}$ or $\frac{2}{3}$

When $\frac{m}{n} = \frac{1}{2}$ i.e. $n = 2m \Rightarrow l = 5m - 3n = -m$ or $\frac{l}{m} = -1$

thus $\frac{m}{n} = \frac{1}{2}$ and $\frac{l}{m} = -1$ giving $\frac{l}{-1} = \frac{m}{1} = \frac{n}{2}$

$$\text{or } \frac{l}{-1} = \frac{m}{1} = \frac{n}{2} = \frac{\sqrt{(l^2 + m^2 + n^2)}}{\sqrt{\{(-1)^2 + 1^2 + 2^2\}}} = \frac{1}{\sqrt{6}}$$

So, direction cosines of one line are $\frac{-1}{\sqrt{6}}, \frac{1}{\sqrt{6}}, \frac{2}{\sqrt{6}}$.

Again when $\frac{m}{n} = \frac{2}{3}$

$$\Rightarrow \frac{l}{m} = \frac{1}{2} \text{ giving } \frac{l}{1} = \frac{m}{2} = \frac{n}{3} = \frac{1}{\sqrt{1^2 + 2^2 + 3^2}} = \frac{1}{\sqrt{14}}$$

$\therefore$ The direction cosines of the other line are $\frac{1}{\sqrt{14}}, \frac{2}{\sqrt{14}}, \frac{3}{\sqrt{14}}$

Q. **Find the direction ratios and direction cosines of the line joining the points $A(6,-7,-1)$ and $B(2,-3,1)$.**

Sol. Direction ratios of AB are $(4,-4,-2) = (2,-2,-1)$

$a^2 + b^2 + c^2 = 9$

Direction cosines are $\left(\pm\frac{2}{3}, \mp\frac{2}{3}, \mp\frac{1}{3}\right)$

Q. **Find direction cosines of a line perpendicular to two lines whose direction ratios are $1,2,3$ and $-2,1,4$.**

Sol. Vector along lines can be taken as $\vec{a} = \hat{i} + 2\hat{j} + 3\hat{k}, \vec{b} = -2\hat{i} + \hat{j} + 4\hat{k}$

$\therefore \quad \vec{a} \times \vec{b} = 5\left(\hat{i} - 2\hat{j} + \hat{k}\right)$

DC's of line $= \left(\frac{1}{\sqrt{6}}, \frac{-2}{\sqrt{6}}, \frac{1}{\sqrt{6}}\right)$ or $\left(\frac{-1}{\sqrt{6}}, \frac{2}{\sqrt{6}}, \frac{-1}{\sqrt{6}}\right)$

Q. **Find the projection of the line segment joining the points $(-1,0,3)$ and $(2,5,1)$ on the line whose direction ratios are 6, 2, 3 .**

Sol. The direction cosines ℓ, m, n of the line are given by

$$\frac{\ell}{6} = \frac{m}{2} = \frac{n}{3} = \frac{\sqrt{\ell^2 + m^2 + n^2}}{\sqrt{6^2 + 2^2 + 3^2}} = \frac{1}{\sqrt{49}} = \frac{1}{7}$$

$$\ell = \frac{6}{7}, m = \frac{2}{7}, n = \frac{3}{7}$$

$\therefore$ The required projection is given by

$$= \left|\ell(x_2 - x_1) + m(y_2 - y_1) + n(z_2 - z_1)\right|$$

$$\left|\frac{6}{7}(2 - (-1)) + \frac{2}{7}(5 - 0) + \frac{3}{7}(1 - 3)\right|$$

$$\left|\frac{6}{7} \times 3 + \frac{2}{7} \times 5 + \frac{3}{7} \times (-2)\right|$$

$$\left|\frac{18}{7} + \frac{10}{7} - \frac{6}{7}\right| = \left|\frac{18 + 10 - 6}{7}\right| = \frac{22}{7}$$

Q. **A variable line has dc's l, m, n and $l + \delta l$, m + δm, n + δn in two adjacent positions. If $\delta\theta$ be the angle between the lines in these two positions then prove that $(\delta\theta)^2 = (\delta l)^2 + (\delta m)^2 + (\delta n)^2$**

Sol. Direction cosines are l, m, n & (l + δl), (m + δm), (n + δn)

Hence $\ell^2 + m^2 + n^2 = 1$ and $(\ell + \delta\ell)^2 + (m + \delta m)^2 + (n + \delta n)^2 = 1$

$\ell^2 + (\delta\ell)^2 + 2\ell\delta\ell + m^2 + (\delta m)^2 + 2m\delta m + n^2 + (\delta n)^2 + 2n\delta n = 1$

$(\delta\ell)^2 + 2\ell d\ell + (\delta m)^2 + 2m\delta m + (\delta n)^2 + 2n\delta n = 1 - (\ell^2 + m^2 + n^2) = 1 - 1 = 0$

$(\delta\ell)^2 + (\delta m)^2 + (\delta n)^2 = -2(\ell\delta\ell + m\delta m + n\delta n)$

Now $\cos\delta\theta = \ell(\ell + \delta\ell) + m(m + \delta m) + n(n + \delta n)$

$= \ell^2 + m^2 + n^2 + \ell\delta\ell + m\delta m + n\delta n$

$$= 1 + \left(\frac{-1}{2}\right)\left[(\delta\ell)^2 + (\delta m)^2 + (\delta n)^2\right]$$

$$\frac{1}{2}[(d\ell)^2 + (\delta m)^2 + (\delta n)^2] = 1 - \cos\delta\theta = 2\sin^2\frac{\delta\theta}{2}$$

Since $\delta\theta$ is very small, so $\sin\frac{\delta\theta}{2} \approx \frac{\delta\theta}{2}$

We get $\frac{1}{2}\left[(\delta\ell)^2 + (\delta m)^2 + (\delta n)^2\right] = \frac{2(\delta\theta)^2}{4}$

$\Rightarrow (\delta\theta)^2 = (\delta\ell)^2 + (\delta m)^2 + (\delta n)^2$ **Hence proved.**

Q. **Find the direction cosines of a line perpendicular to two lines whose dr's are 1, 2, 3 and −2, 1, 4**

Sol. dr's of one line, 1, 2, 3

dr's of other line, −2, 1, 4

vector along the line whose D.R. are 1, 2, 3 $= \hat{i} + 2\hat{j} + 3\hat{k}$

vector along the line whose D.R. are −2, 1, 4 $= -2\hat{i} + \hat{i} + 4\hat{j}$

vector along the line which is $\perp$ar to both given lines is $5\hat{i} - 10\hat{j} + 5\hat{k}$

So, D.R. are (5, −10, 5)

$\sqrt{5^2 + (-10)^2 + 5^2} = 5\sqrt{6}$

So, D.C. are $\frac{5}{5\sqrt{6}}, \frac{-10}{5\sqrt{6}}, \frac{5}{5\sqrt{6}}$

$\left(\frac{1}{\sqrt{6}}, \frac{-2}{\sqrt{6}}, \frac{1}{\sqrt{6}}\right)$

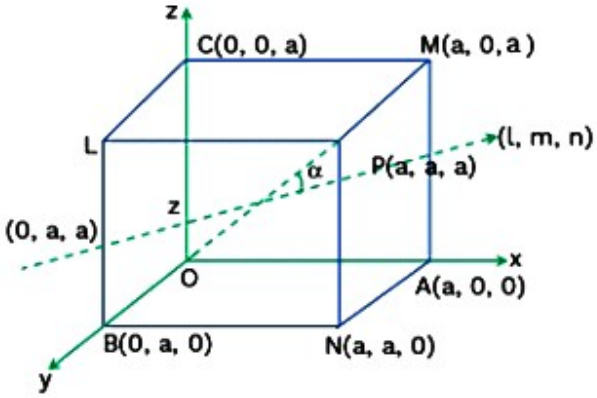
Q. **A line makes angle, α, β, γ and δ with the four diagonals of cube, prove that**

$$\cos^2\alpha+\cos^2\beta+\cos^2\gamma+\cos^2\delta=\frac{4}{3}$$

Sol. Let the cube be shown in the figure, where four diayonals are OP, AL, BM and CN and A(a, 0,0), B(0, a, 0), C(0,0, a), (0, a, a), M(a, 0, a), N(a, a, 0) and P(a, a, a), hence direction cosines of OP are

$$\frac{a}{\sqrt{a^2+a^2+a^2}},\frac{a}{\sqrt{a^2+a^2+a^2}},\frac{a}{\sqrt{a^2+a^2+a^2}}$$

$$=\left(\frac{1}{\sqrt{3}},\frac{1}{\sqrt{3}},\frac{1}{\sqrt{3}}\right)$$



The DC's of AL are $\left(-\frac{1}{\sqrt{3}},\frac{1}{\sqrt{3}},\frac{1}{\sqrt{3}}\right)$

The DC's of BM are $\left(\frac{1}{\sqrt{3}},-\frac{1}{\sqrt{3}},\frac{1}{\sqrt{3}}\right)$

The DC's of CN are $\left(\frac{1}{\sqrt{3}},\frac{1}{\sqrt{3}},-\frac{1}{\sqrt{3}}\right)$

Let the DC's of required line be (l, m, n).

$$\therefore\quad \cos\alpha=\frac{l+m+n}{\sqrt{3}},\cos\beta=\frac{-l+m+n}{\sqrt{3}}$$

$$\cos\gamma=\frac{l-m+n}{\sqrt{3}}\text{ and }\cos\delta=\frac{l+m-n}{\sqrt{3}}$$

$$\cos^2\alpha+\cos^2\beta+\cos^2\gamma+\cos^2\delta$$

$$=\frac{1}{3}\left\{(l+m+n)^2+(-l+m+n)^2+(l-m+z)^2\ +(l+m-n)^2\right\}$$

$$=\frac{4}{3}\left(l^2+m^2+n^2\right)=\frac{4}{3}$$

Planes

General equation of plane:

A linear equation in three variables of the type $ax + by + cz + d = 0$ represents the general equation of a plane.
where a, b, c are not simultaneously zero.

Dividing by d we get $\left(\frac{a}{d}\right)x + \left(\frac{b}{d}\right)y + \left(\frac{c}{d}\right)z + 1 = 0.$

Thus equation of plane involves only three arbitrary constants. Hence in order to determine a unique plane 3 independent conditions are needed.

Division by Coordinate Planes :

The ratios in which the line segment PQ joining $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$ is divided by coordinate planes are as follows.

(i) by yz -plane $: -\frac{x_1}{x_2}$ ratio

(ii) by zx – plane $: -\frac{y_1}{y_2}$ ratio

(iii) by xy – plane $: -\frac{z_1}{z_2}$ ratio

Definition

A plane is a surface such that a line joining any two points on the surface lies completely on it.

Point to Remember!!!

(i) Equation of xy plane is $z = 0$.

(ii) Equation of yz plane is $x = 0$.

(iii) Equation of zx plane is $y = 0$.

Q. **Find the ratio in which the line joining the points $(3, 5, -7)$ and $(-2, 1, 8)$ is divided by yz-plane.**

Sol. Let the line joining the points $(3, 5, -7)$ and $(-2, 1, 8)$ is divided by yz-plane in the ratio $\lambda : 1$, then coordinates of the dividing point will be $\left(\frac{-2\lambda + 3}{\lambda + 1}, \frac{\lambda + 5}{\lambda + 1}, \frac{8\lambda - 7}{\lambda + 1}\right)$

Now above points lies on the yz-plane, so its x-coordinate should be zero i.e.
$\frac{-2\lambda + 3}{\lambda + 1} = 0 \Rightarrow \lambda = \frac{3}{2}$

Hence yz-plane divides line joining the given points in the ratio $\frac{3}{2} : 1$ or $3 : 2$.

DIFFERENT FORMS OF THE EQUATIONS OF PLANES :

1. A point in the plane and a vector normal to it is given :

Let a point A (a) lies in the plane and a vector normal to it is $\vec{n} = a\hat{i} + b\hat{j} + c\hat{k}$.

$P(\vec{r})$ is a moving point whose locus is plane then for every position of vector $\overline{AP}$, vector $\vec{n}$ will be perpendicular to it.

$\therefore \overline{AP} \cdot \vec{n} = 0$

$\Rightarrow (\vec{r} - \vec{a}) \cdot \vec{n} = 0 \Rightarrow \vec{r} \cdot \vec{n} = \vec{a} \cdot \vec{n}$

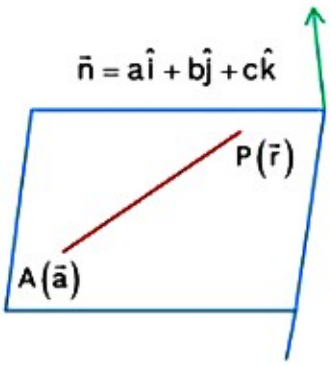
$\Rightarrow \vec{r} \cdot \vec{n} = d$ is general equation of plane in vector form.

It is also known as equation of plane in dot (or scalar) product form.

If $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ and $\vec{a} = x_0\hat{i} + y_0\hat{j} + z_0\hat{k}$ the equation of plane will be

$a(x - x_0) + b(y - y_0) + c(z - z_0) = 0$

This is equation of plane containing point (x_0, y_0, z_0) and perpendicular to vector $a\hat{i} + b\hat{j} + c\hat{k}$



Point to Remember!!!

If equation of a plane is ax + by + cz + d = 0 then a, b, c are direction ratio of normal to the plane.

Plane passing through three given points :

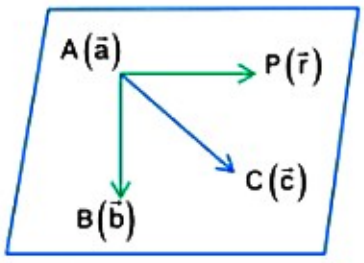
Let three points $A(\vec{a}), B(\vec{b})$ and $C(\vec{c})$ lies in the plane and point $P(\vec{r})$ is moving point whose locus is plane.

$\because \overline{AP}, \overline{AB}$ and $\overline{AC}$ are coplanar.

$\therefore [(\vec{r} - \vec{a} \quad \vec{b} - \vec{a} \quad \vec{c} - \vec{a}] = 0$

represents equation of plane passing through three points.

If $A = (x_1, y_1, z_1), B = (x_2, y_2, z_2), C = (x_3, y_3, z_3)$ and P(x, y, z) then equation of plane is



$$\begin{vmatrix} x-x_1 & y-y_1 & z-z_1 \\ x_2-x_1 & y_2-y_1 & z_2-z_1 \\ x_3-x_1 & y_3-y_1 & z_3-z_1 \end{vmatrix} = 0$$

Plane containing two intersecting lines :

Let the equations of two lines are $\vec{r} = \vec{a} + \lambda\vec{p}$ and $\vec{r} = \vec{a} + \mu\vec{q}$.

Now, $n = \vec{p} \times \vec{q}$ is a vector perpendicular to the plane.

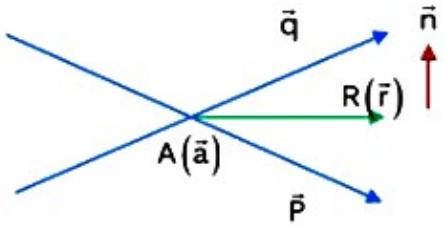
Hence equation of plane is $(\vec{r} - \vec{a}) \cdot (\vec{p} \times \vec{q}) = 0$

$\Rightarrow [\vec{r} - \vec{a}\ \ \vec{p}\ \vec{q}] = 0 \Rightarrow [\vec{r}\ \ \vec{p}\ \ \vec{q}] = [\vec{a}\ \ \vec{p}\ \ \vec{q}]$

Since vectors $\vec{r} - \vec{a}, \vec{p}, \vec{q}$ are coplanar.

Therefore $\vec{r} - \vec{a} = \alpha\vec{p} + \beta\vec{q} \Rightarrow \vec{r} = \vec{a} + \alpha\vec{p} + \beta\vec{q}$

It represents equation of a plane containing point $\vec{a}$ and parallel to two non-collinear vectors $\vec{p}$ and $\vec{q}$. This is also known as parametric equation.



Q. **Express the equation of a plane $\vec{r} = \hat{i} - 2\hat{j} + \lambda(2\hat{i} - \hat{j} + 3\hat{k}) + \mu(3\hat{i} + 4\hat{j} - \hat{k})$ in**

(a) cartesian form.

(b) Scalar product form.

Sol. **(a)** Clearly plane is passing through the point (1, –2, 0) and parallel to vectors $2\hat{i} - \hat{j} + 3\hat{k}$ and $3\hat{i} + 4\hat{j} - \hat{k}$

$\therefore$ Equation of plane is $\begin{vmatrix} x-1 & y+2 & z-0 \\ 2 & -1 & 3 \\ 3 & 4 & -1 \end{vmatrix} = 0$

$\Rightarrow \quad (x-1)(-11) - (y+2)(-11) + z(11) = 0$

$\Rightarrow \quad x - 1 - y - 2 - z = 0 \Rightarrow x - y - z = 3 \Rightarrow x(1) + y(-1) + z(-1) = 3$

(b) Therefore equation of plane is scalar product form in $\vec{r} \cdot (\hat{i} - \hat{j} - \hat{k}) = 3$

Equation of plane containing two parallel lines :

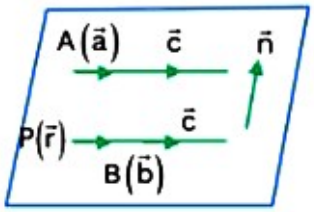
Let lines be $\vec{r} = \vec{a} + \lambda\vec{b}$ and $\vec{r} = \vec{c} + \mu\vec{b}$

vector normal to plane is

$\vec{n} = (\vec{a} - \vec{c}) \times \vec{b}$

$\therefore$ equation of plane is

$(\vec{r} - \vec{a}) \cdot (\vec{a} - \vec{c}) \times \vec{b} = 0$



Alternatively :

Vectors $\vec{r} - \vec{a}, \vec{c} - \vec{a}$ and $\vec{b}$ are coplanar

$\therefore$ equation of plane is $\left[\vec{r} - \vec{a} \quad \vec{c} - \vec{a} \quad \vec{b}\right] = 0$

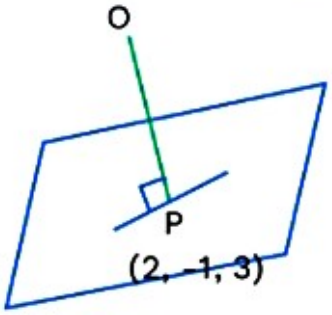
Q. **Find the equation of the plane passing through the point (2, –1, 3) which is the foot of the perpendicular drawn from the origin to the plane.**

Sol. The direction ratios of the normal to the plane are 2, –1, 3

The equation of required plane is

$2(x - 2) - 1(y + 1) + 3(z - 3) = 0$

$\Rightarrow 2x - y + 3z - 14 = 0$



Q. **Find the equation of the plane through $(2,3,-4), (1,-1,3)$ and parallel to x-axis.**

Sol. The equation of the plane passing through $(2,3,-4)$ is

$a(x-2) + b(y-3) + c(z+4) = 0$ ... (1)

Since $(1,-1,3)$ lies on it, we have $a + 4b - 7c = 0$ ... (2)

Since required plane is parallel to x -axis i.e. perpendicular to YZ plane i.e.

$1 \cdot a + 0 \cdot b + 0 \cdot c = 0 \Rightarrow a = 0 \Rightarrow 4b - 7c = 0 \Rightarrow \frac{b}{7} = \frac{c}{4}$

$\therefore$ Equation of required plane is $7y + 4z = 5$.

Q. **Two planes are given by equations $x + 2y - 3z = 0$ and $2x + y + z + 3 = 0$. Find**

(i) DC 's of their normals and the acute angle between them.

(ii) DC 's of their line of intersection.

(iii) Equation of the plane perpendicular to both of them through the point $(2,2,1)$

Sol. $\vec{n}_1 = \hat{i} + 2\hat{j} - 3\hat{k}, \quad \vec{n}_2 = 2\hat{i} + \hat{j} + \hat{k}$

(i) $\cos\theta = \dfrac{\vec{n}_1 \cdot \vec{n}_2}{|\vec{n}_1||\vec{n}_2|} = \dfrac{1}{2\sqrt{21}}$

$\Rightarrow \theta = \cos^{-1}\dfrac{1}{2\sqrt{21}}$ & D.C of their normals are $\left(\pm\dfrac{1}{\sqrt{14}}, \pm\dfrac{2}{\sqrt{14}}, \mp\dfrac{3}{\sqrt{14}}\right)$

& $\left(\pm\dfrac{2}{\sqrt{6}}, \pm\dfrac{1}{\sqrt{6}}, \pm\dfrac{1}{\sqrt{6}}\right)$

(ii) $\vec{n}_1 \times \vec{n}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 2 & -3 \\ 2 & 1 & 1 \end{vmatrix} = 5\hat{i} - 7\hat{j} - 3\hat{k}$

DC's of line of intersection of the plane $\left(\pm\dfrac{5}{\sqrt{83}}, \mp\dfrac{7}{\sqrt{83}}, \mp\dfrac{3}{\sqrt{83}}\right)$

(iii) $(\vec{r} - (2\hat{i} + 2\hat{j} + \hat{k})) \times (\vec{n}_1 \times \vec{n}_2) = 0$

$\Rightarrow 5(x-2) - 7(y-2) - 3(z-1) = 0$

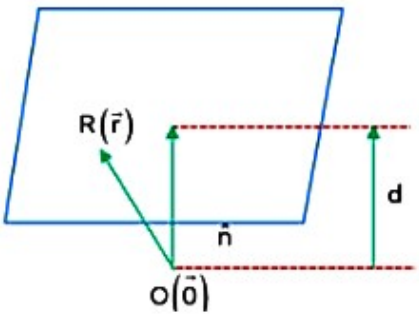
$\Rightarrow 5x - 7y - 3z + 7 = 0$

Normal form of the plane :

A unit vector $\hat{n}$ normal to the plane from origin is known and perpendicular distance of the plane from the origin is d.

Projection of $\vec{r}$ on $\hat{n} = d$

$\vec{r} \cdot \hat{n} = d$ ... (1)



Point to Remember!!!

$d > 0$, as d is distance of the plane from origin. Cartesian form of the plane is $lx + my + nz = d$

where l, m, n are dcs of normal to plane.

Q. **Let equation of plane be $\vec{r}\cdot(6\hat{i}-3\hat{j}-2\hat{k})+1=0$ then find perpendicular distance of plane from origin and also find direction cosines of this perpendicular.**

Sol. Plane is $\vec{r}\cdot(6\hat{i}-3\hat{j}-2\hat{k})=-1\Rightarrow\vec{r}\cdot\left(\frac{-6}{7}\hat{i}+\frac{3}{7}\hat{j}+\frac{2}{7}\hat{k}\right)=\frac{1}{7}$

$\therefore$ perpendicular distance from origin $=\frac{1}{7}$ and dcs of perpendicular $=\left(\frac{-6}{7},\frac{3}{7},\frac{2}{7}\right)$

Q. **Find the vector equation of plane which is at a distance of 8 units from the origin and which is normal to the vector $2\hat{i}+\hat{j}+2\hat{k}$**

Sol. Here, $d=8$ and $\vec{n}=2\hat{i}+\hat{j}+2\hat{k}$

$$\therefore\quad \hat{n}=\frac{\vec{n}}{|\vec{n}|}=\frac{2\hat{i}+\hat{j}+2\hat{k}}{\sqrt{2^2+1^2+2^2}}=\frac{2\hat{i}+\hat{j}+2\hat{k}}{3}$$

Hence, the required equation of plane is,

$\vec{r}\cdot\hat{n}=d$

$$\Rightarrow\vec{r}\cdot\left(\frac{2\hat{i}+\hat{j}+2\hat{k}}{3}\right)=8$$

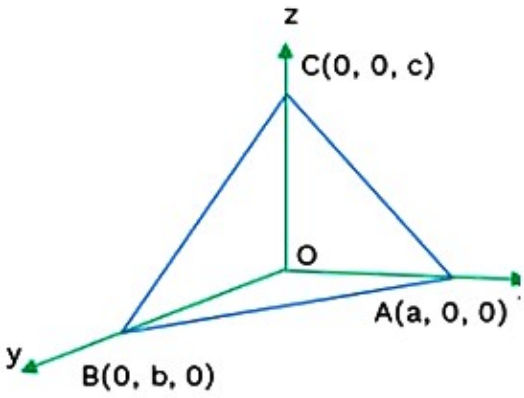
or $\vec{r}\cdot(2\hat{i}+\hat{j}+2\hat{k})=24$

Intercept form the plane :

Equation of plane in the intercept form is

$$\frac{x}{a}+\frac{y}{b}+\frac{z}{c}=1$$

where a = x-intercept,
b = y-intercept,
c = z-intercept



Proof :

Equation of plane passing through three points $A(a,0,0), B(0,\ b,0)$ and $C(0,0,c)$ will be

$$\begin{vmatrix} x-a & y-0 & z-0 \\ -a & b & 0 \\ -a & 0 & c \end{vmatrix}=0$$

$$\Rightarrow(x-a)bc-y(-ac-0)+z(0+ab)=0$$

$\Rightarrow xbc + yac + zab = abc$

$\Rightarrow \frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$

Note :

Area of $\Delta ABC = \frac{1}{2}|\overline{AB} \times \overline{BC}| = \frac{1}{2}|(b\hat{j} - a\hat{i}) \times (ck - b\hat{j})| = \frac{1}{2}|bc\hat{i} + ac\hat{j} + ab\hat{k}|$

$= \frac{1}{2}\sqrt{a^2b^2 + b^2c^2 + c^2a^2} = \sqrt{\left(\frac{ab}{2}\right)^2 + \left(\frac{bc}{2}\right)^2 + \left(\frac{ca}{2}\right)^2}$

$\therefore$ Area of $\Delta ABC = \sqrt{(\text{area of } \Delta OAB)^2 + (\text{area of } \Delta OBC)^2 + (\text{area of } \Delta OCA)^2}$

Examples of Different forms of Plane

Q. **Find the equation of the plane through the point (2, –3, 1) and parallel to the plane 3x – 4y + 2z = 5.**

Sol. Equation of plane parallel to the plane $3x - 4y + 2z = 5$ is $3x - 4y + 2z + \lambda = 0$
It passes through the point (2, –3, 1)
then $3 \times 2 + 4 \times 3 + 2 \times 1 + \lambda = 0$
$\Rightarrow \lambda = -20$
then plane: $3x - 4y + 2z - 20 = 0$

Q. **The intercept made by the plane $\vec{r} \cdot \vec{n} = q$ on the x-axis is**

(A) $\frac{q}{\hat{i} \cdot \vec{n}}$ **(B)** $\frac{\hat{i} \cdot \vec{n}}{q}$ **(C)** $(\hat{i} \cdot \vec{n})q$ **(D)** $\frac{q}{|\vec{n}|}$

Sol. **(A)**

$\vec{r} \cdot \vec{n} = q$

If equation of plane is $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$

then a, b, c are respectively X intercept, Y intercept, Z intercept.
q = coordinates intercept . $\vec{n}$

Hence, X intercept = $\frac{q}{\hat{i} \cdot \vec{n}}$

Q. **Find the equation of the plane which is parallel to the plane x + 5y – 4z + 5 = 0 and the sum of whose intercepts on the co-ordinate axes is 19 units.**

Sol. Equation of plane parallel to the plane $x + 5y - 4z + 5 = 0$ is $x + 5y - 4z + \lambda = 0$
this can be written as $x + 5y - 4z = -\lambda$
$-x - 5y + 4z = \lambda$

$$\frac{x}{\left(\frac{\lambda}{-1}\right)} + \frac{y}{\left(\frac{\lambda}{-5}\right)} + \frac{z}{\left(\frac{\lambda}{4}\right)} = 1$$

x intercept : $-\lambda$

y intercept: $-\frac{\lambda}{5}$

z intercept: $\frac{\lambda}{4}$

sum of intercept s = 19

$$\Rightarrow -\lambda - \frac{\lambda}{5} + \frac{\lambda}{4} = 19$$

$$\Rightarrow \frac{-20\lambda - 4\lambda + 5\lambda}{20} = 19 \Rightarrow \frac{-19\lambda}{20} = 19 \Rightarrow \lambda = -20$$

then plane: $x + 5y - 4z - 20 = 0$

Q. **If from the point P (f, g, h) perpendiculars PL, PM be drawn to yz and zx planes then the equation to the plane OLM is**

(A) $\frac{x}{f} + \frac{y}{g} - \frac{z}{h} = 0$ **(B)** $\frac{x}{f} + \frac{y}{g} + \frac{z}{h} = 0$ **(C)** $\frac{x}{f} - \frac{y}{g} + \frac{z}{h} = 0$ **(D)** $-\frac{x}{f} + \frac{y}{g} + \frac{z}{h} = 0$

Sol. **(A)**

P(f, g, h)
yz plane x = 0 i.e. L = (0, g, h)
zx plane y = 0 i.e. M = (f, 0, h)
O = (0, 0, 0)
Therefore, the equation of required plane is given by,

$$\begin{vmatrix} x-0 & y-0 & z-0 \\ 0-0 & g-0 & h-0 \\ f-0 & 0-0 & h-0 \end{vmatrix} = 0$$

$$\begin{vmatrix} x & y & z \\ 0 & g & h \\ f & 0 & h \end{vmatrix} = 0$$

$\Rightarrow xgh + yfh - zfg = 0$

$\Rightarrow \frac{x}{f} + \frac{y}{g} - \frac{z}{h} = 0$

Q. **The feet of normal from origin on a plane is α, β and γ. Find the equation of the plane.**

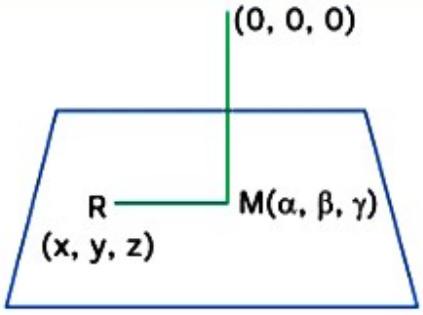
Sol. $\bar{n} = \alpha\hat{i} + \beta\hat{j} + r\hat{k}$

$(\vec{R} - \bar{m})\cdot\bar{n} = 0$

$(\bar{R} - \bar{m})\cdot(\alpha\hat{i} + \beta\hat{j} + r\hat{k}) = 0$

$[(x-\alpha)\hat{i} + (y-\beta)\hat{j} + (z-\gamma)\hat{k}]\cdot(\alpha\hat{i} + \beta\hat{j} + \gamma\hat{k}) = 0$

$\alpha(x-\alpha) + \beta(y-\beta) + \gamma(z-\gamma) = 0$



Q. **Consider three vectors $\vec{p} = \hat{i} + \hat{j} + \hat{k}$, $\vec{q} = 2\hat{i} + 4\hat{j} - \hat{k}$ and $\vec{r} = \hat{i} + \hat{j} + 3\hat{k}$. If $\vec{p}$, $\vec{q}$ and $\vec{r}$ denotes the position vector of three non-collinear points then the equation of the plane containing these points is**

(A) $2x - 3y + 1 = 0$ **(B) $x - 3y + 2z = 0$**

(C) $3x - y + z - 3 = 0$ **(D) $3x - y - 2 = 0$**

Sol. **(D)**

$\overrightarrow{OP} = \vec{p} = \hat{i} + \hat{j} + \hat{k}, \overrightarrow{OQ} = \vec{q} = 2\hat{i} + 4\hat{j} - \hat{k}, \overrightarrow{OR} = \vec{r} = \hat{i} + \hat{j} + 3\hat{k}$

$$\text{Equation of plane} = \begin{vmatrix} x-1 & y-1 & z-1 \\ 1 & 3 & -2 \\ 0 & 0 & 2 \end{vmatrix} = 0$$

$= 2(3(x-1) - 1(y-1)) = 0$

$\Rightarrow 3x - 3 - y + 1 = 0$

$\Rightarrow 3x - y - 2 = 0$

Q. **The cartesian equation of the plane $r = (1 + \lambda - \mu)\hat{i} + (2 - \lambda)\hat{j} + (3 - 2\lambda + 2\mu)\hat{k}$, is**

(A) $2x + y = 5$ **(B) $2x - y = 5$** **(C) $2x + z = 5$** **(D) $2x - z = 5$**

Sol. **(C)**

We have,

$\vec{r} = (1 + \lambda - \mu)\hat{i} + (2 - \lambda)\hat{j} + (3 - 2\lambda + 2\mu)\hat{k}$

$\Rightarrow \vec{r} = (\hat{i} + 2\hat{j} + 3\hat{k}) + \lambda(\hat{i} - \hat{j} - 2\hat{k}) + \mu(-\hat{i} + 2\hat{k})$

Which is a plane passing through $\vec{a} = \hat{i} + 2\hat{j} + 3\hat{k}$ and parallel to the vectors $\vec{b} = \hat{i} - \hat{j} - 2\hat{k}$ and $\vec{c} = -\hat{i} + 2\hat{k}$.

Therefore, it is normal to the vector $\vec{n} = \vec{b} \times \vec{c} = -2\hat{i} - \hat{k}$

Hence, its vector equation is

$(\vec{r} - \vec{a}) \cdot \vec{n} = 0$

$\Rightarrow \quad \vec{r} \cdot \vec{n} = \vec{a} \cdot \vec{n}$

$\Rightarrow \quad \vec{r} \cdot (-2\hat{i} - \hat{k}) = -2 - 3$

$\Rightarrow \quad \vec{r} \cdot (2\hat{i} + \hat{k}) = 5$

So, the cartesian equation of the plane is

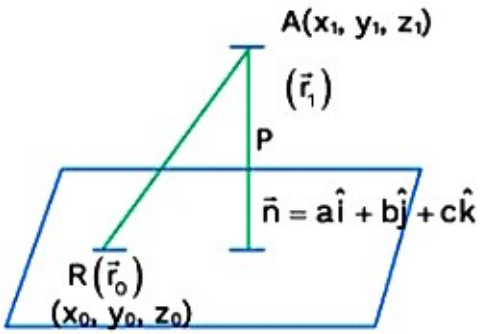
$(x\hat{i} + y\hat{j} + z\hat{k}) \cdot (2\hat{i} + \hat{k}) = 5 \Rightarrow 2x + z = 5$

PERPENDICULAR DISTANCE OF A POINT FROM A PLANE :

Let equation of plane is $\vec{r} \cdot \vec{n} = d$ then perpendicular distance from the point $A(\vec{r}_1)$ on the plane = projection of $\overrightarrow{RA}$ on $\vec{n}$.

$$\Rightarrow P = \frac{\left|(\vec{r}_1 - \vec{r}_0) \cdot \vec{n}\right|}{|\vec{n}|} = \frac{\left|\vec{r}_1 \cdot \vec{n} - \vec{r}_0 \cdot \vec{n}\right|}{|\vec{n}|}$$

$$\Rightarrow P = \frac{\left|\vec{r}_1 \cdot \vec{n} - d\right|}{|\vec{n}|}$$



If equation of plane is $ax + by + cz + d = 0$ then perpendicular distance from point (x_1, y_1, z_1) is given by $\dfrac{\left|ax_1 + by_1 + cz_1 + d\right|}{\sqrt{a^2 + b^2 + c^2}}$

Note :

(i) Planes $a_1x + b_1y + c_1z + d_1 = 0$ and $a_2x + b_2y + c_2z + d_2 = 0$ are

(a) parallel but not identical if $\dfrac{a_1}{a_2} = \dfrac{b_1}{b_2} = \dfrac{c_1}{c_2} \neq \dfrac{d_1}{d_2}$

(b) perpendicular if $a_1a_2 + b_1b_2 + c_1c_2 = 0$

(c) identical if $\dfrac{a_1}{a_2} = \dfrac{b_1}{b_2} = \dfrac{c_1}{c_2} = \dfrac{d_1}{d_2}$

Point to Remember!!!

(i) The equation of a plane parallel to the plane $ax + by + cz + d$ is $ax + by + cz + k = 0$, where k is an arbitrary constant and is determined by the given condition.

(ii) Distance between two parallel planes $ax + by + cz + d_1 = 0$ and $ax + by + cz + d_2 = 0$ is equal to $\dfrac{\left|d_1 - d_2\right|}{\sqrt{a^2 + b^2 + c^2}}$

(ii) 3 planes $a_r x + b_r y + c_r z = d_r \;\; r = 1,2,3$

(a) Can intersect at a point ≡ system of equations in 3 variables having unique solution.

(b) Can intersect coaxially ≡ system of equations in 3 variables having infinite solutions.

(c) May not have a common point ≡ system of equations in 3 variables having no solution.

Q. **A variable plane passes through a fixed point (α,β,γ) and meets the axes in A,B,C. Show that the locus of the point of intersection of the planes through A,B and C parallel to the co-ordinate planes is $\alpha x^{-1} + \beta y^{-1} + \gamma z^{-1} = 1$.**

Sol. Let the equation of the variable plane be $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$ ... (1)

where a, b, c are parameters

The plane (1) passes through the point (α,β,γ).

$\therefore \frac{\alpha}{a} + \frac{\beta}{b} + \frac{\gamma}{c} = 1$ ...(2)

The plane (1) meets the co-ordinate axes in the points A,B and C whose co-ordinates are respectively given by $(a,0,0), (0,b,0)$ and $(0,0,c)$. The equations of the planes through A,B and C are $x = a$, $y = b$, $z = c$ respectively ... (3)

The locus of the point of intersection of these planes is obtained by eliminating the parameters a, b, c between the equation (2), (3). Putting the values of a,b,c from (3) in (2), the required locus is given by $\frac{\alpha}{x} + \frac{\beta}{y} + \frac{\gamma}{z} = 1$ or

$\alpha x^{-1} + \beta y^{-1} + \gamma z^{-1} = 1$

ANGLE BETWEEN TWO PLANES

1. Vector form :

The angle between the two planes is defined as the angle between their normals. Let θ be the angle between planes;

$\vec{r} \cdot \overline{n_1} = d_1$ and $\vec{r} \cdot \overline{n_2} = d_2$ is given by $\cos\theta = \frac{\vec{n}_1 \cdot \vec{n}_2}{|\vec{n}_1 \| \vec{n}_2|}$

2. Cartesian form :

The angle θ between the planes $a_1 x + b_1 y + c_1 z + d_1 = 0$ and $a_2 x + b_2 y + c_2 z + d_2 = 0$ is given by $\cos\theta = \frac{a_1 a_2 + b_1 b_2 + c_1 c_2}{\sqrt{a_1^2 + b_1^2 + c_1^2}\sqrt{a_2^2 + b_2^2 + c_2^2}}$

Angle between a line and a plane :

The angle between a line and a plane is the complement of the angle between the line and the normal to the plane

If α, β, γ be the direction ratios of the line and $ax + by + cz + d = 0$ be the equation of plane and θ be the angle between the line and the plane.

$$\cos(90 - \theta) = \frac{a\alpha + b\beta + c\gamma}{\sqrt{a^2 + b^2 + c^2}\sqrt{\alpha^2 + \beta^2 + \gamma^2}}$$

$$\therefore \sin\theta = \frac{a\alpha + b\beta + c\gamma}{\sqrt{a^2 + b^2 + c^2}\sqrt{\alpha^2 + \beta^2 + \gamma^2}}$$

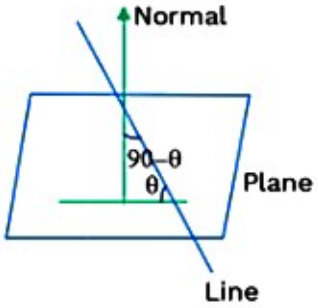
Vector form :

If θ is the angle between the line; $\vec{r} = \vec{a} + \lambda\vec{b}$ and plane $\vec{r} \cdot \vec{n} = d$

$$\sin\theta = \frac{\vec{b} \cdot \vec{n}}{|\vec{b}||\vec{n}|}$$

Point to Remember!!!

Two planes are perpendicular iff $a_1a_2 + b_1b_2 + c_1c_2 = 0$ and Parallel if $\vec{n}_1 \times \vec{n}_2 = 0$



Q. **Find the angle between the planes $2x - y + z = 11$ and $x + y + 2z = 3$.**

Sol. $$\cos\theta = \frac{a_1a_2 + b_1b_2 + c_1c_2}{\sqrt{(a_1^2 + b_1^2 + c_1^2)(a_2^2 + b_2^2 + c_2^2)}}$$

$$\Rightarrow \cos\theta = \frac{2 \cdot 1 + (-1) \cdot 1 + 1 \cdot 2}{\sqrt{2^2 + (-1)^2 + 1^2}\sqrt{1^2 + 1^2 + 2^2}} = \frac{1}{2} \Rightarrow \theta = \frac{\pi}{3}$$

Q. **Find the equation of plane passing through the intersection of planes $2x - 4y + 3z + 5 = 0$, $x + y - z = 6$ and parallel to straight line having direction cosines $(1, -1, -1)$.**

Sol. Equation of required plane be $(2x - 4y + 3z + 5) + \lambda(x + y - z - 6) = 0$

i.e. $(2 + \lambda)x + (-4 + \lambda)y + z(3 - \lambda) + (5 - 6\lambda) = 0$

This plane is parallel to a straight line. So, $al + bm + cn = 0$

$1(2+\lambda)+(-1)(-4+\lambda)+(-1)(3-\lambda)=0$ i.e. $\lambda=-3$

$\therefore$ Equation of required plane is $-x-7y+6z+23=0$.

i.e. $x+7y-6z-23=0$.

Q. **Find the angle between the line $\frac{x+1}{3}=\frac{y-1}{2}=\frac{z-2}{4}$ and the plane**

$2x+y-3z+4=0$

Sol. The given line is parallel to the vector $\vec{b}=3\hat{i}+2\hat{j}+4\hat{k}$ and the given plane is normal to the vector $\vec{n}=2\hat{i}+\hat{j}-3\hat{k}$

$$\sin\theta=\frac{\vec{b}\cdot\vec{n}}{|\vec{b}||\vec{n}|}=\frac{(3\hat{i}+2\hat{j}+4\hat{k})\cdot(2\hat{i}+\hat{j}-3\hat{k})}{\sqrt{3^2+2^2+4^2}\sqrt{2^2+1^2+3^2}}$$

$$=\frac{6+2-12}{\sqrt{29}\sqrt{14}}=\frac{-4}{\sqrt{406}}\quad\therefore\quad\theta=\sin^{-1}\left(\frac{-4}{\sqrt{406}}\right)$$

Q. **Find the equation of plane passing through (2, 2, 0) and (0, 3, 7) and parallel to y-axis.**

Sol. $\overline{OP}=2\hat{i}+2\hat{j}+0\hat{k}$

$\overline{OQ}=0\hat{i}+3\hat{j}+7\hat{k}$

$\overline{PQ}=-2\hat{j}+\hat{j}+7\hat{k}$

$\vec{n}\perp$ y-axis (j)

$\vec{n}\perp\overline{PQ}$

$\vec{n}=\hat{j}\times\overline{PQ}$

$$\vec{n}=\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\0&1&0\\-2&1&7\end{vmatrix}=7\hat{i}+2\hat{k}$$

$\vec{R}=x\hat{i}+y\hat{j}+z\hat{k}$

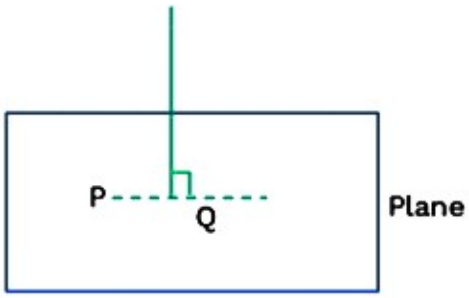
$\overline{PR}=(x-2)\hat{i}+(y-2)\hat{j}+z\hat{k}$

Plane: $\overline{PR}\cdot\vec{n}=0$

$(x-2)7+z\times2=0$

$7x-14+2z=0$

$\boxed{7x+2z-14=0}$



Q. **If the plane $2x - 3y + 6z - 11 = 0$ makes on angle $\sin^{-1}(k)$ with x-axis, then k is equal to**

(A) $\sqrt{3}/2$ **(B) 2/7** **(C)** $\sqrt{2}/3$ **(D) 1**

Sol. **(B)**

$P : 2x - 3y + 6z - 11 = 0$

$\vec{n} = 2\hat{i} - 3\hat{j} + 6\hat{k}$ (normal vector)

$L : \vec{a} = \hat{i}$ represents vector along x-axis. If α is angle between line and plane then

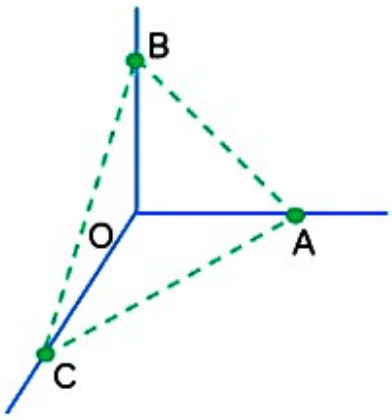
$$\cos(90 - \alpha) = \frac{\vec{n} \cdot \hat{i}}{|\vec{n}||\hat{i}|} = \frac{2}{7}$$

$$\sin\alpha = \frac{2}{7} \Rightarrow \alpha = \sin^{-1}\left(\frac{2}{7}\right)$$

Comparing with given angle in question k = =2/7

Q. **A variable plane is at a constant distance p from the origin and meets the coordinate axes in points A, B and C respectively. Through these points, planes are drawn parallel to the coordinates planes. Find the locus of their point of intersection.**

Sol. Let us say that the line perpendicular to the plane is making α, β, γ with x, y and z axis respectively



So $OB = \frac{p}{\cos\beta} = p\sec\beta = y_1$

$OA = p\sec\alpha = n_1$

$OC = p\sec\gamma = z_1$

We know that $\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$

$$\frac{p^2}{x_1^2} + \frac{p^2}{y_1^2} + \frac{p^2}{z_1^2} = 1$$

$$\Rightarrow \boxed{\frac{1}{x^2} + \frac{1}{y^2} + \frac{1}{z^2} = \frac{1}{p^2}}$$

Q. **Find the equation of the plane parallel to $2x - 6y + 3z = 0$ and at a distance of 2 from the point (1, 2, −3).**

Sol. Equation of plane parallel to the plane $2x - 6y + 3z = 0$ is $2x - 6y + 3z + \lambda = 0$

Length of perpendicular from (1, 2, −3) to $2x - 6y + 3z + \lambda = 0$ is 2 units

$$\left|\frac{2 \times 1 - 6 \times 2 + 3(-3) + \lambda}{\sqrt{4 + 36 + 9}}\right| = 2$$

$$\left|\frac{\lambda - 19}{7}\right| = 2 \quad \Rightarrow \quad \lambda - 19 = \pm 14$$

$$\lambda = +14 + 19 \text{ or } \lambda = 19 - 14$$
$$\lambda = 33 \text{ or } \lambda = 5$$

$2x - 6y + 3z + 33 = 0$
and $2x - 6y + 3z + 5 = 0$

Q. **A plane which always remains at a constant distance p from the origin cuts the co-ordinate axes at A, B, C. Find the locus of**
(a) Centroid of the plane face ABC
(b) Centre of the tetrahedron OABC

Ans. **(a)** Suppose equation of plane is $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$ A(a, 0, 0), B(0, b, 0), C(0, 0, c)

Length of $\perp$ from origin to this plane is p

So $p = \left|\frac{-1}{\sqrt{\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2}}}\right| \Rightarrow \frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2} = \frac{1}{p^2}$ ...(i)

and centroid is $G : \left(\frac{a}{3}, \frac{b}{3}, \frac{c}{3}\right) = (x, y, z)$

$$\Rightarrow a = 3x,\ b = 3y,\ c = 3z$$

replace term in (i)

(i) $\frac{1}{x^2} + \frac{1}{y^2} + \frac{1}{z^2} = \frac{1}{p^2} \Rightarrow \boxed{\frac{1}{x^2} + \frac{1}{y^2} + \frac{1}{z^2} = \frac{9}{p^2}}$

(ii) centre of the tetrahedron

$C : \left(\frac{a}{4}, \frac{b}{4}, \frac{c}{4}\right) = (x, y, z)$

$\Rightarrow a = 4x,\ b = 4y,\ c = 4z$

replace term in (i),

$$\boxed{\frac{1}{x^2} + \frac{1}{y^2} + \frac{1}{z^2} = \frac{16}{p^2}}$$

Two intersecting lines determine a unique plane:

Let the equations of the two lines are $\left.\begin{matrix}\vec{r} = \vec{a} + \lambda\vec{b} \\ \vec{r} = \vec{a} + \mu\vec{c}\end{matrix}\right]$ where $\vec{a}$ is position vector of common point.

Hence the equation of the plane is,

$(\vec{r}-\vec{a})\cdot(\vec{b}\times\vec{c})=0$ or $[\vec{r}-\vec{a}\ \vec{b}\ \vec{c}]=0$ or

$[\vec{r}\vec{b}\vec{c}]=[\vec{a}\vec{b}\vec{c}]$

Condition for coplanarity of two lines :

Let lines are $\vec{r}=\vec{a}+\lambda\vec{b}$ and $\vec{r}=\vec{c}+\mu\vec{d}$

($\vec{b}$ and $\vec{d}$ are non collinear)

$(\vec{r}-\vec{a})\cdot(\vec{b}\times\vec{d})=0$ is the equation of the plane

Condition is $[\vec{a}\vec{b}\vec{d}]=[\vec{c}\vec{b}\vec{d}]$.

Plane containing two parallel lines :

Let lines are $\vec{r}=\vec{a}+\lambda\vec{c}$ and $\vec{r}=\vec{b}+\mu\vec{c}$

$\therefore(\vec{r}-a)\cdot\{(\vec{a}-\vec{b})\times\vec{c}\}=0$

or $[\vec{r}\vec{a}\vec{c}]+[\vec{r}\vec{c}\vec{b}]+[\vec{a}\vec{b}\vec{c}]=0$

or $\left[\vec{r}-\vec{a}\ \ \vec{a}-\vec{b}\ \ \vec{c}\right]=0$

Q. **Find whether the two lines intersect or not. If they do, find the equation of the plane containing them. If they don't find the shortest distance between them. Determine whether the following pair of lines intersect or not.**

(i) $\vec{r}=\hat{i}-\hat{j}+\lambda(2\hat{i}+\hat{k});\vec{r}=2\hat{i}-\hat{j}+\mu(\hat{i}+\hat{j}-\hat{k})$

(ii) $\vec{r}=\hat{i}+\hat{j}-\hat{k}+\lambda(3\hat{i}-\hat{j});\vec{r}=4\hat{i}-\hat{k}+\mu(2\hat{i}+3\hat{k})$

(iii) $\vec{r}=i+2\hat{j}+3\hat{k}+\lambda(2\hat{j}+3\hat{j}+2\hat{k})$ **&** $\vec{r}=2\hat{i}+4\hat{j}+5\hat{k}+\mu(4\hat{i}+6\hat{j}+4\hat{k})$

Sol. **(i)** Here $\overline{a_1}=\hat{i}-\hat{j},\overline{a_2}=2\hat{i}-\hat{j},\vec{b}_1=2\hat{i}+\hat{k}$ and $\overline{b_2}=\hat{i}+\hat{j}-\hat{k}$

$$\text{Now }\left[\overline{a_2}-\overline{a_1}\ \ \overline{b_1}\ \ \overline{b_2}\right]=\begin{vmatrix}2-1 & -1+1 & 0\\ 2 & 0 & 1\\ 1 & 1 & -1\end{vmatrix}$$

$$=\begin{vmatrix}1 & 0 & 0\\ 2 & 0 & 1\\ 1 & 1 & -1\end{vmatrix}$$

$$=-1\neq 0$$

Thus, the two given lines do not intersect.

(ii) Here $\vec{a}_1 = \hat{i} + \hat{j} - \hat{k}, \vec{a_2} = 4\hat{i} - \hat{k}, \vec{b}_1 = 3\hat{i} - \hat{j}$ and $\vec{b_2} = 2\hat{i} + 3\hat{k}$

$$\Rightarrow \left[\vec{a_2} - \vec{a_1}\ \vec{b_1}\ \vec{b_2}\right] = \begin{vmatrix} 4-1 & 0-1 & -1+1 \\ 3 & -1 & 0 \\ 2 & 0 & 3 \end{vmatrix} = \begin{vmatrix} 3 & -1 & 0 \\ 3 & -1 & 0 \\ 2 & 0 & 3 \end{vmatrix} = 0$$

Thus, the two given lines intersect. Let us obtain the point of intersection of the two given lines. For some values of λ and μ, the two values of $\vec{r}$ -must coincide.

Thus, $\hat{i} + \hat{j} - \hat{k} + \lambda(3\hat{i} - \hat{j}) = 4\hat{i} - \hat{k} + \mu(2\hat{i} + 3\hat{k})$

$\Rightarrow (3 + 2\mu - 3\lambda)\hat{i} + (\lambda - 1)\hat{j} + 3\mu\hat{k} = 0$

$\Rightarrow 3 + 2\mu - 3\lambda = 0, \lambda - 1 = 0, 3\mu = 0$

Therefore, the point of intersection is $\vec{r} = 4\hat{i} - \hat{k}$ (by putting $\mu = 0$ in the second equation

(iii) $\vec{r} = \hat{i} + 2\hat{j} + 3\hat{k} + \lambda(2\hat{i} + 3\hat{j} + 2\hat{k})$ & $\vec{r} = 2\hat{i} + 4\hat{j} + 5\hat{k} + \mu(4\hat{i} + 6\hat{j} + 4\hat{k})$

These two lines are parallel and parallel to the vector $\vec{c} = 2\hat{i} + 3\hat{j} + 2\hat{k}$

also $\vec{a} = \hat{i} + 2\hat{j} + 3\hat{k}$ & $\vec{b} = 2\hat{i} + 4\hat{j} + 5\hat{k}$

Distance between these two lines is

$$\frac{|\vec{c} \times (\vec{b} - \vec{a})|}{|\vec{c}|} = \frac{\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 3 & 2 \\ 1 & 2 & 2 \end{vmatrix}}{\sqrt{17}} = \frac{|2\hat{i} - 2\hat{j} + \hat{k}|}{\sqrt{17}} = \frac{3}{\sqrt{17}}$$

Equation of plane containing these two lines.

$[\vec{r} - \vec{a}\ \ \vec{b} - \vec{a}\ \ \vec{c}] = 0$

$$\begin{vmatrix} x-1 & y-2 & z-3 \\ 1 & 2 & 2 \\ 2 & 3 & 2 \end{vmatrix} = 0$$

$\Rightarrow$ $(-2)(x - 1) - (y - 2)(-2) + (z - 3)(-1) = 0$

$\Rightarrow$ $-2x + 2y - z + 1 = 0$

$\Rightarrow$ $2x - 2y + z - 1 = 0$

Line of intersection / Common Line between two intersecting planes :

$\Pi_1 \equiv a_1x + b_1y + c_1z + d_1 = 0$

$\Pi_2 \equiv a_2x + b_2y + c_2z + d_2 = 0$

then line is

$a_1x + b_1y + c_1z + d_1 = a_2x + b_2y + c_2z + d_2 = 0$

Q. **Reduce the equation of line $x - y + 2z = 5$ and $3x + y + z = 6$ in symmetrical form.**

Or

Find the line of intersection of plane $x - y + 2z = 5$ and $3x + y + z = 6$

Sol. Given $x - y + 2z = 5$, $3x + y + z = 6$

Let $z = \lambda$

Then, $x - y = 5 - 2\lambda$

and $3x + y = 6 - \lambda$.

Solving these two equations, $4x = 11 - 3\lambda$

and $4y = 4x - 20 + 8\lambda = -9 + 5\lambda$

The equation of the line is $\frac{4x-11}{-3} = \frac{4y+9}{5} = \frac{z-0}{1}$

Previous Year's Question

If the lines $x = ay + b$, $z = cy + d$ and $x = a'z + b'$, $y = c'z + d'$ are perpendicular, then

[JEE Main 2019]

(A) $cc' + a + a' = 0$ (B) $aa' + c + c' = 0$ (C) $ab' + bc' + 1 = 0$ (D) $bb' + cc' + 1 = 0$

Previous Year's Question

Let L_1 and L_2 be the following straight lines. $L_1 : \frac{x-1}{1} = \frac{y}{-1} = \frac{z-1}{3}$ and $L_2 : \frac{x-1}{-3} = \frac{y}{-1} = \frac{z-1}{1}$.

Suppose the straight line $L : \frac{x-\alpha}{l} = \frac{y-1}{m} = \frac{z-\gamma}{-2}$ lies in the plane containing L_1 and L_2, and passes through the point of intersection of L_1 and L_2. If the line L bisects the acute angle between the lines L_1 and L_2, then which of the following statements is/are TRUE?

[JEE Advanced 2020]

(A) $\alpha - \gamma = 3$ (B) $l + m = 2$ (C) $\alpha - \gamma = 1$ (D) $l + m = 0$

EQUATION OF PLANES BISECTING THE ANGLES BETWEEN TWO GIVEN PLANES :

Cartesian Form :

The equation of the planes bisecting the angles between the planes $a_1x + b_1y + c_1z + d_1 = 0$ and $a_2x + b_2y + c_2z + d_2 = 0$ are

$$\frac{(a_1x + b_1y + c_1z + d_1)}{\sqrt{a_1^2 + b_1^2 + c_1^2}} = \pm \frac{(a_2x + b_2y + c_2z + d_2)}{\sqrt{a_2^2 + b_2^2 + c_2^2}}$$

Know the facts



If angle between bisector plane and one of the plane is less than 45° then it is acute angle bisector otherwise it is obtuse angle bisector.

Vector Form :

The equation of the planes bisecting the angles between the planes $\vec{r} \cdot \vec{n}_1 = d_1$ and $\vec{r} \cdot \vec{n}_2 = d_2$ are

$$\frac{|\vec{r} \cdot \vec{n}_1 - d_1|}{|\vec{n}_1|} = \frac{|\vec{r} \cdot \vec{n}_2 - d_2|}{|\vec{n}_2|} \text{ or } \frac{\vec{r} \cdot \vec{n}_1 - d_1}{|\vec{n}_1|} = \pm \frac{(\vec{r} \cdot \vec{n}_2 - d_2)}{|\vec{n}_2|}$$

$$\text{or } \vec{r} \times (\hat{n}_1 \pm \hat{n}_2) = \frac{d_1}{|\vec{n}_1|} \pm \frac{d_2}{|\vec{n}_2|}$$

FAMILY OF PLANES :

The equation of a plane passing through the lines of intersection of $a_1x + b_1y + c_1z + d_1 = 0$ and $a_2x + b_2y + c_2z + d_2 = 0$ is

$(a_1x + b_1y + c_1z + d_1) + \lambda(a_2x + b_2y + c_2z + d_2) = 0$,

where λ is a constant.

Vectorially :

Equation of a plane passing through the line of intersection of planes $\vec{r} \cdot \vec{n}_1 = d_1$ and $\vec{r} \cdot \vec{n}_2 = d_2$ is given by $(\vec{r} \cdot \vec{n}_1 - d_1) + \lambda(\vec{r} \cdot \vec{n}_2 - d_2) = 0$

Q. **Find the equation of plane containing the line of intersection of the plane $x + y + z - 6 = 0$ and $2x + 3y + 4z + 5 = 0$ and passing through (1, 1, 1)**

Sol. The equation of the plane through the line of intersection of the given planes is,

$(x + y + z - 6) + \lambda(2x + 3y + 4z + 5) = 0$ ... (1)

If it is passes through $(1,1,1)$

$\Rightarrow (1+1+1-6) + \lambda(2+3+4+5) = 0 \Rightarrow \lambda = \frac{3}{14}$

Putting $\lambda = 3/14$ in (i) we get

$(x + y + z - 6) + \frac{3}{14}(2x + 3y + 4z + 5) = 0$

$\Rightarrow 20x + 23y + 26z - 69 = 0$

Q. **Find the equation of the plane passing through (3, 4, −1), which is parallel to the plane** $\vec{r}\cdot(2\hat{i} - 3\hat{j} + 5\hat{k}) + 7 = 0$

Sol. The equation of plane which is parallel to

$\vec{r}\cdot(2\hat{i} - 3\hat{j} + 5\hat{k}) + 7 = 0$ is

$\vec{r}\cdot(2\hat{i} - 3\hat{j} + 5\hat{k}) + \lambda = 0$ ...(i)

or $2x - 3y + 5z + \lambda = 0$

Further (i) will pass through (3, 4, − 1)

If (2) (3) + (−3) (4) + 5(−1) + λ = 0

or $-11 + \lambda = 0 \Rightarrow \lambda = 11$

Thus, equation of the required plane is $\vec{r}\cdot(2\hat{i} - 3\hat{j} + 5\hat{k}) + 11 = 0$

Q. **The plane x − y − z = 4 is rotated through 90° about its line of intersection with the plane**
x + y + 2z = 4. Find its equation in the new position.

Sol. Given planes are $x - y - z = 4$ ...(i)

& $x + y + 2z = 4$ ...(ii)

Equation of any plane passing through the line of Intersection of these planes,

$(x - y - z - 4) + \lambda(x + y + 2z - 4) = 0$

$x(1 + \lambda) + y(-1 + \lambda) + z(-1 + 2\lambda) + (-4 - 4\lambda) = 0$ ...(A)

The direction cosine of a normal to plane (A) are

$$\frac{1+\lambda}{\sqrt{6\lambda^2 - 4\lambda + 3}}, \frac{(\lambda - 1)}{\sqrt{6\lambda^2 - 4\lambda + 3}}, \frac{-1 + 2\lambda}{\sqrt{6\lambda^2 - 4\lambda + 3}}$$

Since the angle between plane (A) and (i) is 90°, so we have

$$\frac{(1+\lambda)}{\sqrt{3}\sqrt{6\lambda^2-4\lambda+3}}+\frac{(-1)(\lambda-1)}{\sqrt{3}\sqrt{6\lambda^2-4\lambda+3}}-\frac{(-1)(-1+2\lambda)}{\sqrt{3}\sqrt{6\lambda^2-4\lambda+3}}=0$$

$0 = 1 + \lambda - \lambda + 1 + 1 - 2\lambda \Rightarrow \lambda = +3/2$

So equation of plane is $5x + y + 4z = 20$

Q. **Find the condition of lines $x = ay + b$, $z = cy + d$, $x = a'y + b'$, $z = c'y + d'$ are perpendicular.**

Sol. The equations of straight lines can be rewritten as

$$x = ay + b, z = cy + d \Rightarrow \frac{x-b}{a} = \frac{y-0}{1} = \frac{z-d}{c}$$

$$\text{and } x = a'y + b', z = c'y + d' \Rightarrow \frac{x-b'}{a'} = \frac{y-0}{1} = \frac{z-d'}{c'}$$

The above lines are perpendicular if $aa' + 1.1 + c.c' = 0$.

Q. **Find the equation of the plane containing the line of intersection of the plane $x + y + z - 6 = 0$ and $2x + 3y + 4z + 5 = 0$ and passing through the points (1, 1, 1)**

Sol. The equation of a plane through the line of intersection of the given plane is

$(x + y + z - 6) + \lambda(2x + 3y + 4z + 5) = 0$ ...(i)

If line (1) passes through (1, 1, 1), we have $-3 + 14\lambda = 0$

$$\Rightarrow \quad \lambda = \frac{3}{14}.$$

Putting $\lambda = \frac{3}{14}$ in line (i), we obtain the equation of the required plane as

$$(x + y + z - 6) + \frac{3}{14}(2x + 3y + 4z + 5) = 0$$

$$\Rightarrow \quad 20x + 23y + 26z - 69 = 0$$

Q. **The plane $ax + by = 0$ is rotated through an angle α about its line of intersection with the plane $z = 0$. Show that the equation to the plane is the new position is $ax + by \pm z\sqrt{a^2+b^2}\tan\alpha = 0$**

Sol: Given planes are $ax + by = 0$ ...(i)

and $z = 0$ ...(ii)

Therefore, the equation of any plane passing through the line of intersection of planes

(i) and (ii) may be taken as

$ax + by + kz = 0$ ...(iii)

The direction cosines of a normal to the plane (iii) are

$$\frac{a}{\sqrt{a^2+b^2+k^2}}, \frac{b}{\sqrt{a^2+b^2+k^2}} \text{ and } \frac{k}{\sqrt{a^2+b^2+k^2}}$$

The direction cosines of a normal to the plane (i) are

$$\frac{a}{\sqrt{a^2+b^2}} \frac{b}{\sqrt{a^2+b^2}} \text{ and } 0$$

Since the angle between the planes (i) and (iii) is α,

$$\cos\alpha = \frac{a\cdot a + b\cdot b + k\cdot 0}{\sqrt{a^2+b^2+k^2}\sqrt{a^2+b^2}} = \sqrt{\frac{a^2+b^2}{a^2+b^2+k^2}}$$

$$\Rightarrow k^2\cos^2\alpha = a^2\left(1-\cos^2\alpha\right) + b^2\left(1-\cos^2\alpha\right)$$

$$\Rightarrow k^2 = \frac{\left(a^2+b^2\right)\sin^2\alpha}{\cos^2\alpha} \Rightarrow k = \pm\sqrt{a^2+b^2}\tan\alpha$$

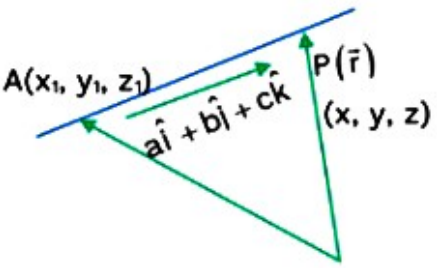
Putting this in (iii), we get the equation of the plane as $ax + by \pm z\sqrt{a^2+b^2}\tan\alpha = 0$

STRAIGHT LINES

Symmetric Form :

1. Equation of a straight line passing through (x_1, y_1, z_1) and having drs as a, b, c is

$$\frac{x-x_1}{a} = \frac{y-y_1}{b} = \frac{z-z_1}{c} = \lambda$$



Proof:

A vector parallel to line will be $a\hat{i} + b\hat{j} + c\hat{k}$

A vector along the line can be written as

$$\overline{AP} = (x-x_1)\hat{i} + (y-y_1)\hat{j} + (z-z_1)\hat{k}$$

$\because$ vector $\overline{AP}$ is parallel to $a\hat{i} + b\hat{j} + c\hat{k}$

$$\therefore \frac{x-x_1}{a} = \frac{y-y_1}{b} = \frac{z-z_1}{c} = \lambda$$

2. Equation of straight line passing through two points (x_1, y_1, z_1) and (x_2, y_2, z_2) will be

$$\frac{x-x_1}{x_2-x_1} = \frac{y-y_1}{y_2-y_1} = \frac{z-z_1}{z_2-z_1}$$

Point to Remember!!!

- Any point on this line can be taken as $(x_1 + \lambda l, y_1 + \lambda m, z_1 + \lambda n)$.
- If dcs of line be l, m, n then its equation will $\frac{x-x_1}{l} = \frac{y-y_1}{m} = \frac{z-z_1}{n} = \lambda$ and any point on this line can be taken as $(x_1 + \lambda a, y_1 + \lambda b, z_1 + \lambda c)$

Note:

(a) Equation of x-axis is $\frac{x}{1}=\frac{y}{0}=\frac{z}{0}$ (or) $y = z = 0$

(b) Equation of y-axis is $\frac{x}{0}=\frac{y}{1}=\frac{z}{0}$ (or) $x = z = 0$

(c) Equation of z-axis $\frac{x}{0}=\frac{y}{0}=\frac{z}{1}$ (or) $x = y = 0$

Here zero in denominator represents that line is perpendicular to that axis.

Ex. Line $\frac{x-2}{3}=\frac{y+1}{-2}$ and $z = 2$ is written as $\frac{x-2}{3}=\frac{y+1}{-2}=\frac{z-2}{0}$

This line is perpendicular to z-axis or parallel to xy plane at a distance of 2 units.

Unsymmetrical form of straight line :

The equations $a_1x+b_1y+c_1z+d_1=0$ and $a_2x+b_2y+c_2z+d_2=0$ together represents a line in unsymmetrical form. This represent equation of line of intersection of planes $a_1x+b_1y+c_1z+d_1=0$ and $a_2x+b_2y+c_2z+d_2=0$

Procedure to convert Unsymmetrical Form of straight line to Symmetrical Form :

Let the direction ratios of the line of intersection (AB) of two planes

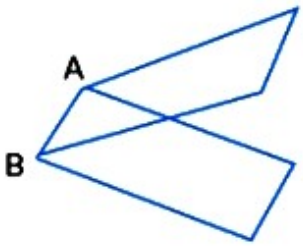
$a_1x+b_1y+c_1z+d_1=0$ ...(1)

and $a_2x+b_2y+c_2z+d_2=0$ ...(2)

are a, b, c

Direction ratios of normal to plane (1) are a_1, b_1, c_1 and

Direction ratios of normal to plane (2) are a_2, b_2, c_2



Line AB lies in both the planes (1) and (2)

hence normals to (1) and (2) are perpendicular to AB.

Hence $aa_1+bb_1+cc_1=0$ and $aa_2+bb_2+cc_2=0$

these two will give the proportional values of a, b, c.

Let the line AB cuts the xy plane at $(x_1, y_1, 0)$

Hence $a_1x_1+b_1y_1=-d_1$ and $a_2x_1+b_2y_1=-d_2$ This will give a point on the line AB

$\therefore$ equation of AB is $\frac{x-x_1}{a}=\frac{y-y_1}{b}=\frac{z-0}{c}$.

Q. **Find the angle between the line** $x-2y+z=0=x+2y-2z$ **and** $x+2y+z=0=3x+9y+5z$

Sol. Let a_1, b_1, c_1 be the direction ratios of the line $x-2y+z=0$ and $x+2y-2z=0$.

Since it lies in both the planes, therefore, it is $\perp$ to the normals to the two planes.

$\therefore a_1 - 2b_1 + c_1 = 0 \Rightarrow a_1 + 2b_1 - 2c_1 = 0$

Solving these two equations by cross-multiplication, we have

$$\frac{a_1}{4-2} = \frac{b_1}{1+2} = \frac{c_1}{2+2} \quad \text{or} \quad \frac{a_1}{2} = \frac{b_1}{3} = \frac{c_1}{4}$$

Let a_2, b_2, c_2 be the direction ratios of the line $x+2y+z=0=3x+9y+5z$.

Then the discussed above

$a_2 + 2b_2 + c_2 = 0 \qquad 3a_2 + 9b_2 + 5c_2 = 0$

$$\Rightarrow \frac{a_2}{10-9} = \frac{b_2}{3-5} = \frac{c_2}{9-6} \quad \text{or} \quad \frac{a_2}{1} = \frac{b_2}{-2} = \frac{c_2}{3}$$

Let θ be the angle between the given lines. Then

$$\cos\theta = \frac{a_1a_2 + b_1b_2 + c_1c_2}{\sqrt{a_1^2 + b_1^2 + c_1^2}\sqrt{a_2^2 + b_2^2 + c_2^2}}$$

$$= \frac{(1)(2) + (-2)(3) + (3)(4)}{\sqrt{2^2 + 3^2 + 4^2}\sqrt{1^2 + (-2)^2 + (3)^2}}$$

$$= \frac{2-6+12}{\sqrt{29}\sqrt{14}} = \frac{8}{\sqrt{406}} \Rightarrow \theta = \cos^{-1}\left(\frac{8}{\sqrt{406}}\right)$$

Q. **Find the coordinates of the point where the line joining the points** $(2,-3,1)$ **and** $(3,-4,-5)$ **cuts the plane** $2x+y+z=7$

Sol. The direction ratios of the line are $3-2, -4-(-3), -5-1$ i.e. $1, -1, -6$

Hence equation of the line joining the given points is $\frac{x-2}{1} = \frac{y+3}{-1} = \frac{z-1}{-6} = r$ (say)

Coordinates of any point on this line are $(r+2, -r-3, -6r+1)$

If this point lies on the given plane $2x+y+z=7$, then

$2(r+2) + (-r-3) + (-6r+1) = 7 \Rightarrow r = -1$

Coordinates of the point are $(-1+2, -(-1)-3, -6(-1)+1)$ i.e. $(1, -2, 7)$.

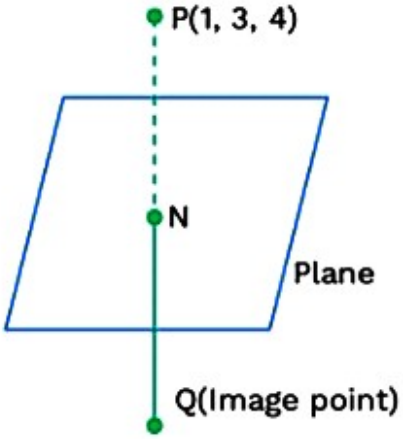
Q. **Find unsymmetrical form the equations of the line**
$3x + 2y - z - 4 = 0 = 4x + y - 2z + 3.$

Sol. The equation of the line in unsymmetrical form are
$3x + 2y - z - 4 = 0$, $4x + y - 2z + 3 = 0$ ...(1)
Let l, m, n be the direction cosines of the line. Since the line is common to both the planes, it is perpendicular to the normals to both the planes. Hence,
$3l + 2m - n = 0, 4l + m - 2n = 0$

Solving these we get, $\frac{l}{-4+1} = \frac{m}{-4+6} = \frac{n}{3-8}$

i.e. $\frac{l}{-3} = \frac{m}{2} = \frac{n}{-5}$ and $\sqrt{(-3)^2 + 2^2 + (-5)^2} = \sqrt{38}$

So, direction cosines of the lines are
$\frac{-3}{\sqrt{38}}, \frac{2}{\sqrt{38}}, \frac{-5}{\sqrt{38}}$



Now to find the coordinates of a point on a line. Let us find out the point where it meets the plane $z = 0$. Putting $z = 0$ in the equation given by (1), we have $3x + 2y - 4 = 0, 4x + y + 3 = 0$. Solving these, we get $x = -2, y = 5$

So, one point of the line is $(-2, 5, 0)$

$\therefore$ Equation of the line in symmetrical form is $\frac{x+2}{-3} = \frac{y-5}{2} = \frac{z-0}{-5}$ i.e.

$\frac{x+2}{-3} = \frac{y-5}{2} = \frac{z}{-5}$.

Q. **Find the equation of the plane which contains the two parallel lines**
$\frac{x+1}{3} = \frac{y-2}{2} = \frac{z}{1}$ **and** $\frac{x-3}{3} = \frac{y+4}{2} = \frac{z-1}{1}$.

Sol. The equations of the two parallel lines are
$\frac{x+1}{3} = \frac{y-2}{2} = \frac{z-0}{1}$ ...(1)

and $\frac{x-3}{3} = \frac{y+4}{2} = \frac{z-1}{1}$ ...(2)

the equation of any plane through the line (1) is
$a(x + 1) + b(y - 2) + cz = 0$ ...(3)

where $3a + 2b + c = 0$ ...(4)

the line (2) will also lie on the plane (3) if the point (3, -4, 1) lies on the plane (3), and for this we have $a(3 + 1) + b(-4 - 2) + c = 0$ or $4a - 6b + c = 0$ ...(5)

Solving (4) and (5), we get $\frac{a}{8} = \frac{b}{1} = \frac{c}{-26}$

Putting the values of a, b, c in (3), the required equation of the plane is $8x + y - 26z + 6 = 0$.

Q. **Find the shortest distance between the lines $\frac{x-3}{3} = \frac{y-8}{-1} = \frac{z-3}{1}, \frac{x+3}{-3} = \frac{y+7}{2} = \frac{z-6}{4}$. Also find the equation of line of shortest distance.**

Sol. Given lines are $\frac{x-3}{3} = \frac{y-8}{-1} = \frac{z-3}{1} = r_1$ (say) ...(1)

$\frac{x+3}{-3} = \frac{y+7}{2} = \frac{z-6}{4} = r_2$ (say) ...(2)

any point on line (1) is $P(3r_1 + 3, 8 - r_1, r_1 + 3)$ and on line (2) is $Q(-3 - 3r_2, 2r_2 - 7, 4r_2 + 6)$.

If PQ is line of shortest distance, then direction ratios of

$PQ = (3r_1 + 3) - (-3 - 3r_2), (8 - r_1) - (2r_2 - 7), (r_1 + 3) - (4r_2 + 6)$

i.e. $3r_1 + 3r_2 + 6, -r_1 - 2r_2 + 15, r_1 - 4r_2 - 3$

As PQ is perpendicular to lines (1) and (2)

$\therefore 3(3r_1 + 3r_2 + 6) - 1(-r_1 - 2r_2 + 15) + 1(r_1 - 4r_2 - 3) = 0$

$\Rightarrow 11r_1 + 7r_2 = 0$ ...(3)

and $-3(3r_1 + 3r_2 + 6) + 2(-r_1 - 2r_2 + 15) + 4(r_1 - 4r_2 - 3) = 0$

i.e. $7r_1 + 29r_2 = 0$ ...(4)

On solving equations (3) and (4), we get $r_1 = r_2 = 0$.

So, point P(3, 8, 3) and Q(-3, -7, 6)

$\therefore$ Length of shortest distance $PQ = \sqrt{\left\{(-3-3)^2 + (-7-8)^2 + (6-3)^2\right\}} = 3\sqrt{30}$.

Direction ratios of shortest distance line is 2, 5, -1.

$\therefore$ Equation of shortest distance line $\frac{x-3}{2} = \frac{y-8}{5} = \frac{z-3}{-1}$.

Q. **Find the distance of the point (1, -2, 3) from the plane $x - y + z = 5$ measured parallel to the line $\frac{x}{2} = \frac{y}{3} = \frac{z}{-6}$.**

Sol. Here we are not to find perpendicular distance of the point from the plane but distance measured along with the given line. The method is as follow:

The equation of the line through the point (1, -2, 3) and parallel to given line is

$\frac{x-1}{2} = \frac{y+2}{3} = \frac{z-3}{-6} = r$ (say)

The coordinate of any point on it is $(2r + 1, 3r - 2, -6r + 3)$.
If this point lies in the given plane then

$2r + 1 - (3r - 2) + (-6r + 3) = 5 \Rightarrow -7r = -1$ or $r = \frac{1}{7}$

$\therefore$ Point of intersection is $\left(\frac{9}{7}, \frac{-11}{7}, \frac{15}{7}\right)$

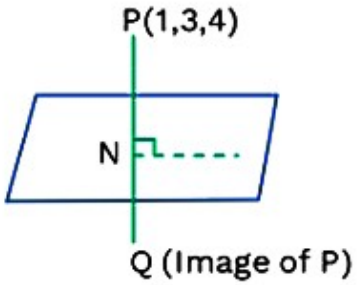
$\therefore$ The required distance = the distance between the points $(1, -2, 3)$ and $\left(\frac{9}{7}, \frac{-11}{7}, \frac{15}{7}\right)$

$$= \sqrt{\left(1-\frac{9}{7}\right)^2 + \left(-2+\frac{11}{7}\right)^2 + \left(3-\frac{15}{7}\right)^2} = \frac{\sqrt{49}}{7} = 1 \text{ Unit.}$$

Q. **Find the image of the point (1, 3, 4) in the plane $2x - y + z + 3 = 0$.**

Sol. As it is clear from the figure that PQ will be perpendicular to the plane and foot of this perpendicular is mid point of PQ i.e. N.
So, direction ratios of line PQ is 2, –1, 1

$\Rightarrow$ Equation of line PQ $= \frac{x-1}{2} = \frac{y-3}{-1} = \frac{z-4}{1} = r$ (say)



$\Rightarrow$ Any point on line PQ is $(2r + 1, -r + 3, r + 4)$
If this point lies on the plane, then
$2(2r + 1) - (-r + 3) + (r + 4) + 3 = 0$
$\Rightarrow r = -1$
$\therefore$ Coordinate of foot of perpendicular N = (–1, 4, 3).
As N is middle point of PQ.

$\therefore -1 = \frac{1+x_1}{2}, 4 = \frac{3+y_1}{2}, 3 = \frac{4+z_1}{2} \Rightarrow x_1 = -3, y_1 = 5, z_1 = 2$

$\therefore$ Image of point P(1, 3, 4) is the point Q(–3, 5, 2).

Q. **Determine whether each statement is true or false**

(a) Two lines parallel to a third line are parallel.
(b) Two lines perpendicular to a third line are parallel.
(c) Two planes parallel to a third plane are parallel.
(d) Two planes perpendicular to a third plane are parallel.
(e) Two lines parallel to a plane are parallel.
(f) Two lines perpendicular to a plane are parallel.
(g) Two planes parallel to a line are parallel.

(h) Two planes perpendicular to a line are parallel.
(i) Two planes either intersect or are parallel.
(j) Two lines either intersect or are parallel.
(k) A plane and a line either intersect or are parallel.

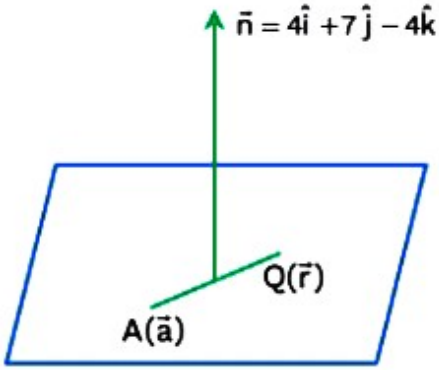
Ans. (a) True, (b) False, (c) True, (d) False, (e) False, (f) True,
(g) False, (h) True, (i) True, (j) False, (k) True

Q. **Find the equation of a plane passing through the point A(3,–2, 1) and perpendicular to the vector $4\hat{i}+7\hat{j}-4\hat{k}$. If PM be perpendicular from the point P(1, 2, –1) to this plane, find its length.**

Sol. Let O be the origin, then $\vec{a} = \overline{OA} = 3\hat{i}-2\hat{j}+\hat{k}$

Let $\vec{n} = 4\hat{i}+7\hat{j}-4\hat{k}$
Let Q(x, y, z) be any point on the plane
Then, $\vec{r} = \overline{OQ} = x\hat{i}+y\hat{j}+z\hat{k}$

Now $\vec{r}-\vec{a} = (x-3)\hat{i}+(y+2)\hat{j}+(z-1)\hat{k}$



Now, equation of the required plane
passing point A($\vec{a}$) and perpendicular to $\vec{n}$ is
$(\vec{r}-\vec{a})\cdot\vec{n} = 0$

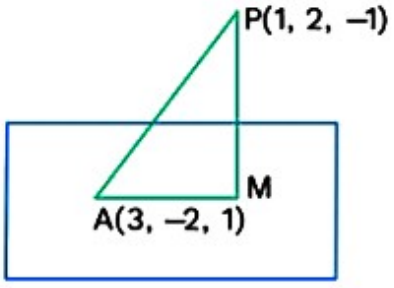
$\therefore\ 4(x-3)+7(y+2)-4(z-1) = 0$
or $4x+7y-4z+6 = 0$. This is the required equation of the plane.
Now PM $\perp$ AM

$\therefore$ Unit vector parallel to $\overline{PM}$

$$\hat{n} = \frac{\vec{n}}{|\vec{n}|} = \frac{4\hat{i}+7\hat{j}-4\hat{k}}{9}$$

Now PM = length of projection of $\overline{PA}$ on $\overline{PM}$

$$= |\overline{PA}\cdot\hat{n}| = \left|(2\hat{i}-4\hat{j}+2\hat{k})\cdot\left(\frac{4\hat{i}+7\hat{j}-4\hat{k}}{9}\right)\right| = \frac{28}{9}$$



Second method:

Let $\vec{\alpha} = \overline{OP} = \hat{i}+2\hat{j}-\hat{k}$

Alternatively, $PM = \left|\frac{(\vec{\alpha}-\vec{a})\cdot\vec{n}}{n}\right|$

Here, $\vec{\alpha}-\vec{a} = \overline{AP} = -2\hat{i}+4\hat{j}-2\hat{k}$ and $\vec{n} = 4\hat{i}+7\hat{j}-4\hat{k}$

$$PM = \left|\frac{-8+28+8}{9}\right| = \frac{28}{9}$$

Q. **Find the shortest distance and the vector equation of the line of shortest distance between the lines given by** $\vec{r} = 3\hat{i} + 8\hat{j} + 3\hat{k} + \lambda(3\hat{i} - \hat{j} + \hat{k})$ **and**

$\vec{r} = -3\hat{i} - 7\hat{j} + 6\hat{k} + \mu(-3\hat{i} + 2\hat{j} + 4\hat{k})$

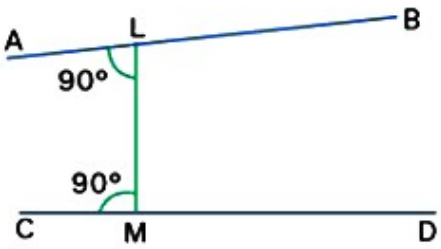
Sol. Given lines are $\vec{r} = 3\hat{i} + 8\hat{j} + 3\hat{k} + \lambda(3\hat{i} - \hat{j} + \hat{k})$ ...(i)

and $\vec{r} = -3\hat{i} - 7\hat{j} + 6\hat{k} + \mu(-3\hat{i} + 2\hat{j} + 4\hat{k})$ ...(ii)

Equation of lines (i) and (ii) in cartesian from are,

$AB : \frac{x-3}{3} = \frac{y-8}{-1} = \frac{z-3}{1} = \lambda$ ...(iii)

and $CD : \frac{x+3}{-3} = \frac{y+7}{2} = \frac{z-6}{4} = \mu$ ...(iv)



Let $L \equiv (3\lambda + 3, -\lambda + 8, \lambda + 3)$

and $M \equiv (-3\mu - 3, 2\mu - 7, 4\mu + 6)$

Direction ratios of LM are

$3\lambda + 3\mu + 6, -\lambda - 2\mu + 15, \lambda - 4\mu - 3$

Since $LM \perp AB$

$\therefore \quad 3(3\lambda + 3\mu + 6) - 1(-\lambda - 2\mu + 15) + 1(\lambda - 4\mu - 3) = 0$

or $11\lambda + 7\mu = 0$ ...(v)

Again $LM \perp CD$

$\therefore \quad -3(3\lambda + 3\mu + 6) + 2(-\lambda - 2\mu + 15) + 4(\lambda - 4\mu - 3) = 0$

or $-7\lambda - 29\mu = 0$ ...(vi)

Solving (v) and (vi), we get $\lambda = 0, \mu = 0$

$\therefore \quad L \equiv (3, 8, 3), M \equiv (-3, -7, 6)$

Hence shortest distance $LM = \sqrt{(3+3)^2 + (8+7)^2 + (3-6)^2} = \sqrt{270} = 3\sqrt{30}$ units

Vector equation of LM is

$\vec{r} = 3\hat{i} + 8\hat{j} + 3\hat{k} + t(6\hat{i} + 15\hat{j} - 3\hat{k})$

Note: Cartesian equation of LM is $\frac{x-3}{6} = \frac{y-8}{15} = \frac{z-3}{-3}$

Miscellaneous Examples on Line and Plane and Shortest Distance between Skew Lines

Q. **Find the vector equation of the following planes in cartesian form:**

$\vec{r} = \hat{i} - \hat{j} + \lambda(\hat{i} + \hat{j} + \hat{k}) + \mu(\hat{i} - 2\hat{j} + 3\hat{k})$

Sol. The equation of the plane $\vec{r} = \hat{i} - \hat{j} + \lambda(\hat{i} + \hat{j} + \hat{k}) + \mu(\hat{i} - 2\hat{j} + 3\hat{k})$

Let $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$

Hence, the equation is $(x\hat{i} + y\hat{j} + z\hat{k}) - (\hat{i} - \hat{j}) = \lambda(\hat{i} + \hat{j} + \hat{k}) + \mu(\hat{i} - 2\hat{j} + 3\hat{k})$

Thus vectors $(x\hat{i}+y\hat{j}+z\hat{k})-(\hat{i}-\hat{j}),\hat{i}+\hat{j}+\hat{k},\hat{i}-2\hat{j}+3\hat{k}$ are coplanar.

Therefore, the equation of the plane is

$$\begin{vmatrix} x-1 & y-(-1) & z-0 \\ 1 & 1 & 1 \\ 1 & -2 & 3 \end{vmatrix} = 0 \text{ or } 5x-2y-3z-7=0$$

Q. **Find the points where line $\frac{x-1}{2}=\frac{y+2}{-1}=\frac{z}{1}$ intersects xy, yz and zx planes.**

Sol. line meets xy-plane where z = 0

Hence from the given equation of line, $\frac{x-1}{2}=\frac{y+2}{-1}=\frac{0}{1}$

$\Rightarrow$ x = 1 and y = – 2.

$\Rightarrow$ Line meets xy-plane at (1, –2, 0).

Line meets yz-plane where x = 0

Hence from the given equation of line, $\frac{0-1}{2}=\frac{y+2}{-1}=\frac{z}{1}$

$\Rightarrow \quad z=\frac{-1}{2}$ and $y=-\frac{3}{2}$

$\Rightarrow$ Line meets yz -plane at $\left(0,-\frac{3}{2},\frac{-1}{2}\right)$

Line meets zx-plane where y = 0

Hence from the given equation of line $\frac{x-1}{2}=\frac{0+2}{-1}=\frac{z}{1}$

$\Rightarrow$ z = –2, x = – 3

$\Rightarrow$ Line meets zx-plane at (–3, 0, –2)

Q. **Find the point where the line $\frac{x-2}{3}=\frac{y-2}{0}=\frac{z-1}{2}$ cuts the plane x + y + z = 0.**

Sol. Given line, $\frac{x-2}{3}=\frac{y-2}{0}=\frac{z-1}{2}=(\lambda)$

Any point on this line $(3\lambda+2, 2, 2\lambda+1)$

Given plane, x + y + z = 0, point lies on this plane,

$3\lambda+2+2+2\lambda+1=0$

$\lambda=-1$

So point is (–1, 2, –1).

Q. **Find equation of line passing through (0, 0, 0) and $\perp$ to fixed line $\frac{x-1}{2} = \frac{y+2}{-3} = \frac{z}{5}$ and || to fixed plane $x - y + z + 2 = 0$.**

Sol. Equation of line passing through (0, 0, 0) and having direction rations $< a, b, c >$ is

$$\frac{x}{a} = \frac{y}{b} = \frac{z}{c} \quad \ldots(i)$$

Line (i) is perpendicular to $\frac{x-1}{2} = \frac{y-2}{-3} = \frac{z}{5}$

So we get, $2a - 3b + 5c = 0$ ...(A)

and parallel to plane $x - y + z + 2 = 0$

So we get $a - b + c = 0$ ...(B)

by (A) & (B) $$\frac{a}{\begin{vmatrix} -3 & 5 \\ -1 & 1 \end{vmatrix}} = \frac{-b}{\begin{vmatrix} 2 & 5 \\ 1 & 1 \end{vmatrix}} = \frac{c}{\begin{vmatrix} 2 & -3 \\ 1 & -1 \end{vmatrix}}$$

$$\Rightarrow \quad \frac{a}{2} = \frac{-b}{-3} = \frac{c}{1} \Rightarrow \frac{a}{2} = \frac{b}{3} = \frac{c}{1}$$

Equation of line (i), $\frac{x}{2} = \frac{y}{3} = \frac{z}{1}$

Q. **Find the length of the perpendicular drawn from point (2, 3, 4) to line $\frac{4-x}{2} = \frac{y}{6} = \frac{1-z}{3}$**

Sol. Let P be the foot of the perpendicular from A(2, 3, 4) to the given line l whose equation is $\frac{4-x}{2} = \frac{y}{6} = \frac{1-z}{3}$ or $\frac{x-4}{-2} = \frac{y}{6} = \frac{z-1}{-3} = k$ (say). Therefore, ...(i)

$x = 4 - 2k, y = 6k, z = 1 - 3k$

As P lies on (i), coordinates of P are

(4 – 2k, 6k, 1 – 3k) from some value of k.

The direction ratios of AP are

(4 – 2k – 2, 6k –3, 1 – 3k – 4)

or (2 – 2k, 6k – 3, –3 – 3k).

Also, the direction ratios of l are –2, 6 and –3.

Since AP $\perp$ l,

$\Rightarrow -2(2 - 2k) + 6(6k - 3) - 3(-3 - 3k) = 0$

$\Rightarrow -4 + 4k + 36k - 18 + 9 + 9k = 0$ or $49k - 13 = 0$ or $k = 13/49$

We have $AP^2 = (4 - 2k - 2)^2 + (6k - 3)^2 + (1 - 3k - 4)^2$

$= (2 - 2k)^2 + (6k - 3)^2 + (-3 - 3k)^2$



$$= 4 - 8k + 4k^2 + 36k^2 - 36k + 9 + 9 + 18k + 9k^2$$

$$= 22 - 26k + 49k^2$$

$$= 22 - 26\left(\frac{13}{49}\right) + 49\left(\frac{13}{49}\right)^2$$

$$= \frac{22 \times 49 - 26 \times 13 + 13^2}{49} = \frac{909}{49}$$

$$AP = \frac{3}{7}\sqrt{101}$$

Previous Year's Question

A line l passing through the origin is perpendicular to the lines **[IIT-Advance 2013]**

$$\ell_1 : (3 + t)\hat{i} + (-1 + 2t)\hat{j} + (4 + 2t)\hat{k}, -\infty < t < \infty$$

$$\ell_2 : (3 + 2s)\hat{i} + (3 + 2s)\hat{j} + (2 + s)\hat{k}, -\infty < s < \infty$$

Then, the coordinate(s) of the point(s) on ℓ_2 at a distance of $\sqrt{17}$ from the point of intersection of ℓ and ℓ_1 is(are) :

(A) $\left(\frac{7}{3}, \frac{7}{3}, \frac{5}{3}\right)$ (B) $(-1, -1, 0)$ (C) $(1, 1, 1)$ (D) $\left(\frac{7}{9}, \frac{7}{9}, \frac{8}{9}\right)$

Q. **Find equation of line passing through (1, 1, 2) and || to planes $x + 2y + 2z + 3 = 0$ and $x + 2y - z + 4 = 0$.**

Sol. Equation of line passing through (1, 1, 2) and having direction ratios < a, b, c > is

$$\frac{x-1}{a} = \frac{y-1}{b} = \frac{z-2}{c} - \text{(i)}$$

This line is parallel to the planes $x + 2y + 2z + 3 = 0$ & $x + 2y - z + 4 = 0$

So we get, $a + 2b + 2c = 0$

& $a + 2b - c = 0$

We get, $\frac{a}{\begin{vmatrix} 2 & 2 \\ 2 & -1 \end{vmatrix}} = \frac{-b}{\begin{vmatrix} 1 & 2 \\ 1 & -1 \end{vmatrix}} = \frac{c}{\begin{vmatrix} 1 & 2 \\ 1 & 2 \end{vmatrix}}$

$$\Rightarrow \frac{a}{-6} = \frac{b}{3} = \frac{c}{0}$$

Hence equation of line, $\frac{x-1}{-6} = \frac{y-1}{3} = \frac{z-2}{0}$

Q. **Reduce the equation of line $x - y + 2z = 5$ and $3x + y + z = 6$ in symmetrical form.** or **Find the line of intersection of planes $x - y + 2z = 5$ and $3x + y + z = 6$.**

Sol. Given $x - y + 2z = 5$, $3x + y + z = 6$.

Let $z = \lambda$.

Then $x - y = 5 - 2\lambda$ and $3x + y = 6 - \lambda$.

Solving these two equations, $4x = 11 - 3\lambda$ and $4y = 4x - 20 + 8\lambda = -9 + 5\lambda$.

The equation of the line is $\frac{4x-11}{-3} = \frac{4y+9}{5} = \frac{z-0}{1}$

Q. **Prove that the lines $3x + 2y + z - 5 = 0 = x + y - 2z - 3$ and $2x - y - z = 0 = 7x + 10y - 8z$ are perpendicular to each other.**

Sol. $3x + 2y + z - 5 = 0 = x + y - 2z - 3$

Direction ratio of line passing through the intersection of these two planes,

$$\bar{n}_1 \times \bar{n}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & 2 & 1 \\ 1 & 1 & -2 \end{vmatrix} = \hat{i}(-5) - \hat{j}(-7) + \hat{k}(1) = -5\hat{i} + 7\hat{j} + \hat{k}$$

$2x - y - z = 0 = 7x + 10y - 8z$

$$\bar{n}_3 \times \bar{n}_4 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & -1 & -1 \\ 7 & 10 & -8 \end{vmatrix} = \hat{i}(18) - \hat{j}(-16 + 7) + \hat{k}(27) = 18\hat{i} + 9\hat{j} + 27\hat{k}$$

Product of these: $-5 \times 18 + 7 \times 9 + 1 \times 27 = 0$

hence perpendicular to each other

Q. **The distance of the point (1, −2, 3) from the plane $x - y + z = 5$ measured parallel to the line $\frac{x}{2} = \frac{y}{3} = \frac{z-1}{-6}$ is**

(A) 1 **(B) 2** **(C) 4** **(D) None of these**

Sol. **(A)**

The equation of the line passing through P(1, − 2, 3) and parallel to the given line is

$$\frac{x-1}{2} = \frac{y+2}{3} = \frac{z-3}{-6}$$

Suppose it meets the plane $x - y + z = 5$ at the point Q given by

$\frac{x-1}{2} = \frac{y+2}{3} = \frac{z-3}{-6}$

$= \lambda$ i.e. $(2\lambda + 1, 3\lambda - 2, -6\lambda + 3)$

This lies on $x - y + z = 5$. Therefore,

$2\lambda + 1 - 3\lambda + 2 - 6\lambda + 3 = 5$

$\Rightarrow \quad -7\lambda = -1 \Rightarrow \lambda = \frac{1}{7}$

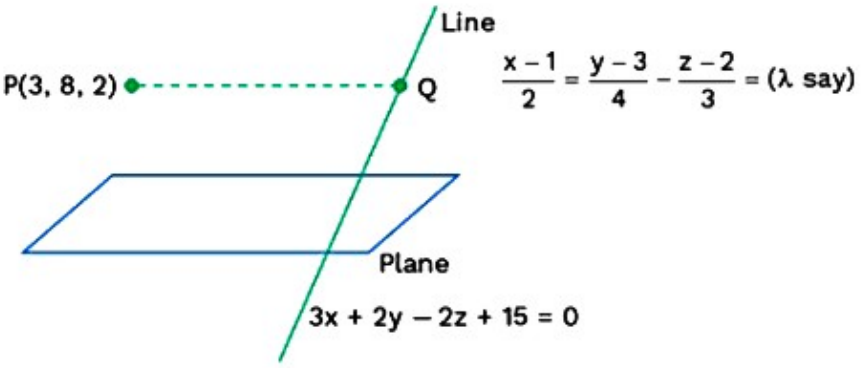
So, the coordinate of Q are $\left(\frac{9}{7}, -\frac{11}{7}, \frac{15}{7}\right)$

Hence, required distance = PQ = $\sqrt{\frac{4}{49} + \frac{9}{49} + \frac{36}{49}} = 1$.

Q. **Find distance of the point A(3, 8, 2) from the line $\frac{x-1}{2} = \frac{y-3}{4} = \frac{z-2}{3}$ measured parallel to the Π : 3x + 2y −2z +15 = 0.**

Sol. Any point 'Q' on the given line: $(2\lambda + 1, 4\lambda + 3, 3\lambda + 2)$

Direction rations of 'PQ'



$(2\lambda - 2, 4\lambda - 5, 3\lambda)$

PQ is parallel to the given plane, hence

$3(2\lambda - 2) + 2(4\lambda - 5) - 2(3\lambda) = 0$

$\Rightarrow 6\lambda - 6 + 8\lambda - 10 - 6\lambda = 0$

$\Rightarrow \lambda = 2$

Hence PQ $\sqrt{4 + 9 + 36} = 7$

Q. **Find the vector equation of a line passing through (2, −1,1) and parallel to the line whose equation is $\frac{x-3}{2} = \frac{y+1}{7} = \frac{z-2}{-3}$.**

Sol. Since the required line is parallel to $\frac{x-3}{2}=\frac{y+1}{7}=\frac{z-2}{-3}$, it follows that the required line passing through $A(2\hat{i}-\hat{j}+\hat{k})$ has the direction of $2\hat{i}+7\hat{j}-3\hat{k}$.

Hence, the vector equation of the required line is $\vec{r}=2\hat{i}-\hat{j}+\hat{k}+\lambda(2\hat{i}+7\hat{j}-3\hat{k})$ where λ is a parameter.

Q. Prove that the three lines from O with direction cosines l_1, m_1, n_1; l_2, m_2, n_2 and l_3, m_3, n_3 are coplanar, if $l_1(m_2n_3 - n_2m_3) + m_1(n_2l_3 - l_2n_3) + n_1(l_2m_3 - l_3m_2) = 0$.

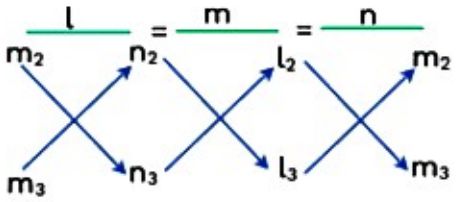
Sol. Here, three given lines are coplanar, if they have common perpendicular.

Let DC's of common perpendicular be l, m and n.

$$ll_1 + mm_1 + nn_1 = 0 \quad ...(i)$$

$$ll_2 + mm_2 + nn_2 = 0 \quad ...(ii)$$

and $$ll_3 + mm_3 + nn_3 = 0 \quad ...(iii)$$



Solving Eqs. (ii) and (iii) by cross multiplication method, we get

$$\Rightarrow \frac{l}{m_2n_3-n_2m_3}=\frac{m}{n_2l_3-n_3l_2}=\frac{n}{l_2m_3-l_3m_2}=k$$

$$\Rightarrow l=k(m_2n_3-n_2m_3), m=k(n_2l_3-n_3l_2), n=k(l_2m_3-l_3m_2)$$

Substituting in Eq. (i), we get

$$\Rightarrow l=k(m_2n_3-n_2m_3), m=k(n_2l_3-n_3l_2), n=k(l_2m_3-l_3m_2)$$

$$\Rightarrow l_1(m_2m_3-n_2m_2)+m_1(n_2l_3-n_3l_2)+n_1(l_2m_3-l_3m_2)$$

Q. Find the shortest distance between the lines $\frac{x-1}{2}=\frac{y-2}{3}=\frac{z-3}{4}$ and $\frac{x-2}{3}=\frac{y-4}{4}=\frac{z-5}{5}$. Also obtain the equation of the lines of the shortest distance.

Sol. (i) The two given lines are $\frac{x-1}{2}=\frac{y-2}{3}=\frac{z-3}{4}=r_1$ (say) ...(i)

and $\frac{x-2}{3}=\frac{y-4}{4}=\frac{z-5}{5}=r_2$ (say) ...(ii)

Any point on (i) is given by $P(2r_1 + 1, 3r_1 + 2, 4r_1 + 3)$

And any point on (ii) is given by $Q(3r_2 + 2, 4r_2 + 4, 5r_2 + 5)$

Direction ratios of PQ are given by $3r_2 - 2r_1 + 1$, $4r_2 - 3r_1 + 2$ and $5r_2 - 4r_1 + 2$

Since PQ is perpendicular to (i), we get

$2(3r_2 - 2r_1 + 1) + 3(4r_2 - 3r_1 + 2) + 4(5r_2 - 4r_1 + 2) = 0$
Or $38r_2 - 29r_1 + 16 = 0$
Also PQ is perpendicular to (ii), we get
$3(3r_2 - 2r_1 + 1) + 4(4r_2 - 3r_1 + 2) + 5(5r_2 - 4r_1 + 2) = 0$

or $50r_2 - 38r_1 + 21 = 0$
Solving (iii) and (iv), we obtain $r_2 = -(1/6)$, $r_1 = (1/3)$.
Therefore, coordinates of P and Q are $\left(\frac{5}{3}, 3, \frac{13}{3}\right)$ and $\left(\frac{3}{2}, \frac{10}{3}, \frac{25}{6}\right)$,

respectively.

Thus, $PQ^2 = \left(\frac{3}{2} - \frac{5}{3}\right)^2 + \left(\frac{10}{3} - 3\right)^2 + \left(\frac{25}{6} - \frac{13}{3}\right)^2 = \left(-\frac{1}{6}\right)^2 + \left(\frac{1}{3}\right)^2 + \left(-\frac{1}{6}\right)^2 = \frac{1}{6}$

$\Rightarrow \quad PQ = \frac{1}{\sqrt{6}}$

The equation of the line of the shortest distance is given by

$$\frac{x - (5/3)}{(3/2) - (5/3)} = \frac{y - 3}{(10/3) - 3} = \frac{z - (13/3)}{(25/6) - (13/3)}$$

$$\frac{x - (5/3)}{-(1/6)} = \frac{y - 3}{(1/3)} = \frac{z - (13/3)}{-(1/6)}$$

$$\frac{x - (5/3)}{1} = \frac{y - 3}{-2} = \frac{z - (13/3)}{1}$$

Alternative method for finding the shortest distance:

Line (i) is passing through the point $(x_1, y_1, z_1) \equiv (1, 2, 3)$ and is parallel to vector

$a_1\hat{i} + b_1\hat{j} + c_1\hat{k} \equiv 2\hat{i} + 3\hat{j} + 4\hat{k}$

Line (ii) is passing through the point $(x_2, y_2, z_2) \equiv (2, 4, 5)$ and is parallel to the vector

$a_2\hat{i} + b_2\hat{j} + c_2\hat{k} \equiv 3\hat{i} + 4\hat{j} + 5\hat{k}$

Hence the shortest distance between the lines using the formula

$$\frac{\begin{vmatrix} x_2 - x_1 & y_2 - y_1 & z_2 - z_1 \\ a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{vmatrix}}{\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{vmatrix}} \text{ is } \frac{\begin{vmatrix} 2-1 & 4-2 & 5-3 \\ 2 & 3 & 4 \\ 3 & 4 & 5 \end{vmatrix}}{\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 3 & 4 \\ 3 & 4 & 5 \end{vmatrix}} = \frac{1}{\sqrt{6}}$$

Q. **Find equation of line through (1, 1, 1) and intersect the lines** $\frac{x-1}{2}=\frac{y-2}{3}=\frac{z-3}{4}, \frac{x+2}{1}=\frac{y-3}{2}=\frac{z+1}{4}$.

Sol. Any line passing through the point (1, 1, 1) is $\frac{x-1}{a}=\frac{y-1}{b}=\frac{z-1}{c}$ ...(i)

This line intersects the line $\frac{x-1}{2}=\frac{y-2}{3}=\frac{z-3}{4}$.

If $a:b:c \neq 2:3:4$ and $\begin{vmatrix} 1-1 & 2-1 & 3-1 \\ a & b & c \\ 2 & 3 & 4 \end{vmatrix}=0$

$\Rightarrow a - 2b + c = 0$

Again, line (i) intersects line $\frac{x-(-2)}{1}=\frac{y-3}{2}=\frac{z-(-1)}{4}$ ...(ii)

If $a:b:c \neq 1:2:4$ and $\begin{vmatrix} -2-1 & 3-1 & -1-1 \\ a & b & c \\ 1 & 2 & 4 \end{vmatrix}=0$

$\Rightarrow 6a + 5b - 4c = 0$

From (ii) and (iii) by cross multiplication, we have $\frac{a}{8-5}=\frac{b}{6+4}=\frac{c}{5+12}$

$\Rightarrow \frac{a}{3}=\frac{b}{10}=\frac{c}{17}$

So, then required line is $\frac{x-1}{3}=\frac{y-1}{10}=\frac{z-1}{17}$

Q. **Prove that the three planes $2x + y - 4z - 17 = 0$, $3x + 2y - 2z - 25 = 0$ and $2x - 4y + 3z + 25 = 0$ intersect at a point and find the coordinate of the points.**

Sol. Given planes,

$2x + y - 4z - 17 = 0$

$3x + 2y - 2z - 25 = 0$

$2x - 4y + 3z + 25 = 0$

$D=\begin{vmatrix} 2 & 1 & -4 \\ 3 & 2 & -2 \\ 2 & -4 & 3 \end{vmatrix}=-4-13+64=47$

$D_1=\begin{vmatrix} 17 & 1 & -4 \\ 25 & 2 & -2 \\ -25 & -4 & 3 \end{vmatrix}=-34+325=141$

$$D_2 = \begin{vmatrix} 2 & 17 & -4 \\ 3 & 25 & -2 \\ 2 & -25 & 3 \end{vmatrix} = 50 - 221 + 500 = 329 \text{ \& similarly } D_3 = -47$$

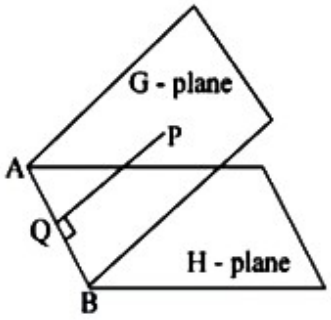
$$x = \frac{D_1}{D}, y = \frac{D_2}{D}, z = \frac{D_3}{D}$$

$$x = \frac{141}{47}, y = \frac{329}{47}, z = -\frac{47}{47}$$

$x = 3, y = 7, z = -1$
hence point is (3, 7, –1).

Line of Greatest Slope

AB is the line of intersection of G-plane and H-plane (horizontal plane). Line of greatest slope on a given plane, drawn through a given point on the plane, is the line through the point 'P' perpendicular to the line of intersection of the given plane with any horizontal plane.



Q. **Assuming the plane (4x – 3y + 7z) = 0 to be horizontal plane, find the equation of line of greatest slope in the plane 2x + y – 5z = 0 and passing through the point (2, 1, 1).**

Sol. The required plane must be perpendicular to the line of intersection of the planes, i.e.
$2x + y - 5z = 0$ and $4x - 3y + 7z = 0$
Let the direction ratio's of the line of intersection of the given planes are a, b, c
Therefore, $2a + b - 5c = 0$
$4a - 3b + 7c = 0$
After solving, $\frac{a}{4} = \frac{b}{17} = \frac{c}{5}$

Now direction ratios of required line be proportional to ℓ, m, n and the line passes through the point (2, 1, 1)

equation of line, $\frac{x-2}{\ell} = \frac{y-1}{m} = \frac{z-1}{n}$ ...(i)

whereas, $2\ell + m - 5n = 0$
$4\ell + 17m + 5n = 0$

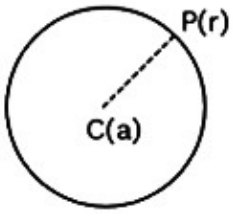
After solving, $\frac{\ell}{3} = \frac{m}{-1} = \frac{n}{1}$ ...(ii)

Hence required line,

$$\frac{x-2}{3} = \frac{y-1}{-1} = \frac{z-1}{1}$$

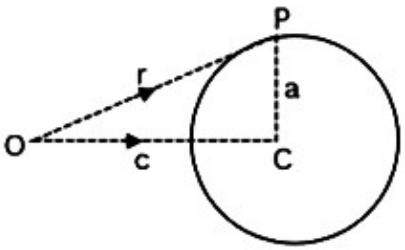
Sphere

A sphere is the locus of a point which moves in space in such a way that its distance from a fixed point always remains constant. The fixed point is called the centre of the sphere and the fixed distance is called the radius of sphere. Shown as in adjoining figure.



Equation of Sphere whose Centre c and Radius is a

Let O be the origin of reference and C be the centre of sphere whose position vector is c. Let P be any point on the surface of the sphere whose position vector is r.



Thus, OP = r and OC = c

$\therefore$ CP = OP − OC = r − c

Now, $|\vec{r} - \vec{c}| = a$ [radius of sphere]

$\Rightarrow |\vec{r} - \vec{c}|^2 = a^2$

$\Rightarrow |\vec{r} - \vec{c}| . |\vec{r} - \vec{c}| = a^2$

$\Rightarrow r^2 - 2\vec{r}.\vec{c} + c^2 = a^2$

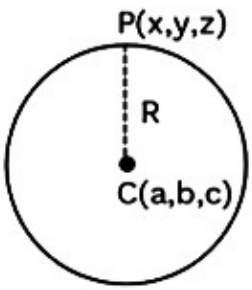
$\Rightarrow r^2 - 2\vec{r}.\vec{c} + (c^2 - a^2) = 0$

which is the required equation of sphere.

Cartesian Equation of a Sphere

The equation of sphere with centre (a, b, c) and radius R is

$(x-a)^2 + (y-b)^2 + (z-c)^2 = R^2$



Proof :

Let C be the centre of the sphere.

Then, coordinates of C are (a, b, c). Let P(x, y, z) be any point on the sphere, then

CP = R

$\Rightarrow$ $CP^2 = R^2$

$\Rightarrow$ $(x-a)^2 + (y-b)^2 + (z-c)^2 = R^2$

Since, P(x, y, z) is an arbitrary point on the sphere, therefore required equation of the sphere is

$(x-a)^2 + (y-b)^2 + (z-c)^2 = R^2$

Remarks

1. The above equation is called the central form of a sphere. If the centre is at the origin, then equation of sphere is, $x^2 + y^2 + z^2 = R^2$ (known as the standard form of the sphere)
2. Above equation can also be written as

$x^2 + y^2 + z^2 - 2ax - 2by - 2cz + (a^2 + b^2 + c^2 - R^2) = 0$

which has the following characteristics of the equation of sphere

(i) It is a second degree equation in x, y and z.

(ii) The coefficient of x^2, y^2 and z^2 are all equal.

(iii) The term containing the product of x y, y z and z x are absent.

Q. **Find the equation of sphere whose centre is (5, 2, 3) and radius is 2 in cartesian form.**

Sol. The required equation of the sphere is

$(x-5)^2 + (y-2)^2 + (z-3)^2 = 2^2$

$\Rightarrow x^2 + y^2 + z^2 - 10x - 4y - 6z + 34 = 0$

General Equation of Sphere

The equation $x^2 + y^2 + z^2 + 2ux + 2vy + 2wz + d = 0$ represents a sphere with centre $(-u, -v, -w)$ i.e.

$\left(-\frac{1}{2}\text{ coefficient of } x, -\frac{1}{2}\text{ coefficient of } y, -\frac{1}{2}\text{ coefficient of } z\right)$

and radius $= \sqrt{u^2 + v^2 + w^2 - d}$

Point to Remember!!!

The equation $x^2 + y^2 + z^2 + 2ux + 2vy + 2wz + d = 0$ represents a real sphere. If $u^2 + v^2 + w^2 - d > 0$. If $u^2 + v^2 + w^2 - d = 0$, then it represents a point sphere. The sphere is imaginary, if $u^2 + v^2 + w^2 - d < 0$.

Q. **Find the centre and radius of the sphere $2x^2 + 2y^2 + 2z^2 - 2x - 4y + 2z + 3 = 0$**

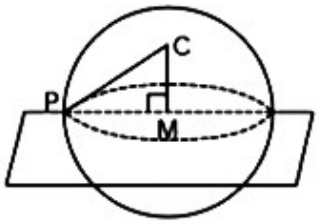
Sol. The given equation

$$x^2 + y^2 + z^2 - x - 2y + z + \frac{3}{2} = 0$$

where centre is

$$\left(-\frac{1}{2}\text{ coefficient of } x, -\frac{1}{2}\text{ coefficient of } y, -\frac{1}{2}\text{ coefficient of } z\right)$$

$$\therefore \text{ Centre} = \left(\frac{1}{2}, 1, -\frac{1}{2}\right)$$



$$\text{and Radius} = \sqrt{\left(\frac{1}{2}\right)^2 + (1)^2 + \left(-\frac{1}{2}\right)^2 - \frac{3}{2}}$$

$$= \sqrt{\frac{1}{4} + 1 + \frac{1}{4} - \frac{3}{2}} = 0$$

$\therefore$ Given sphere represents a point sphere $\left(\frac{1}{2}, 1, -\frac{1}{2}\right)$.

Q. **Find the sphere through (1, 0, 0), (0, 1, 0), (0, 0, 1) having radius as small as possible.**

Sol. Let the equation of the required sphere be

$$x^2 + y^2 + z^2 + 2ux + 2vy + 2wz + d = 0 \quad \ldots(i)$$

As the sphere passes through (1, 0, 0), (0, 1, 0) and (0, 0, 1), we get
$1 + 2u + d = 0$, $1 + 2v + d = 0$ and $1 + 2w + d = 0$

$$\Rightarrow \quad u = v = w = -\frac{1}{2}(d + 1)$$

If R is the radius of the sphere, then $R^2 = u^2 + v^2 + w^2 - d$

$$\Rightarrow \quad R^2 = \frac{3}{4}(d + 1)^2 - d$$

$$= \frac{3}{4}\left[d^2 + 2d + 1 - \frac{4}{3}d\right]$$

$$= \frac{3}{4}\left[d^2 + \frac{2}{3}d + 1\right]$$

$$= \frac{3}{4}\left[\left(d + \frac{1}{3}\right)^2 + 1 - \frac{1}{9}\right]$$

$$= \frac{3}{4}\left[\left(d + \frac{1}{3}\right)^2 + \frac{8}{9}\right]$$

The last equation shows that R^2 (and thus R) will be the least if an only if

$d = -\frac{1}{3}$

Therefore, $u = v = w = -\frac{1}{2}\left(1 - \frac{1}{3}\right) = -\frac{1}{3}$

Hence, the equation of the required sphere is

$x^2 + y^2 + z^2 - \frac{2}{3}(x + y + z) - \frac{1}{3} = 0$ or $3\left(x^2 + y^2 + z^2\right) - 2(x + y + z) - 1 = 0$.

Diameter Form of the Equation of a Sphere

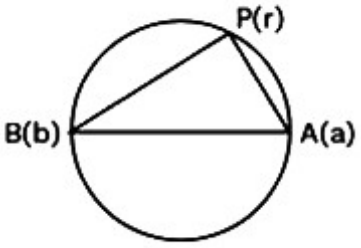
If the position vectors of the extremities of a diameter of a sphere are $\vec{a}$ and $\vec{b}$, then its equation is

$$\left(\vec{r} - \vec{a}\right) . \left(\vec{r} - \vec{b}\right) = 0$$

$$\Rightarrow \quad |r|^2 - \vec{r} . \left(\vec{a} + \vec{b}\right) + \vec{a} . \vec{b} = 0$$

Proof :

Let a and b be the position vectors of the extremities A and B of a diameter A B of sphere. Let r be the position vector of any point P on the sphere. Then,



$AP = \vec{r} - \vec{a}$ and $BP = \vec{r} - \vec{b}$

Since, the diameter of a sphere subtends a right at any point on the sphere, therefore

$$\Rightarrow \quad \angle APB = \frac{\pi}{2}$$

$$\Rightarrow \quad \overline{AP} \cdot \overline{BP} = 0$$

$$\Rightarrow \quad (\vec{r} - \vec{a}).(\vec{r} - \vec{b}) -= 0$$

$$\vec{r}.\vec{r} - \vec{r}.\vec{b} - \vec{r}.\vec{a} + \vec{a}.\vec{b} = 0$$

$$|\vec{r}|^2 - (\vec{a} + \vec{b}).\vec{r} + \vec{a}.\vec{b} = 0$$

This is the required equation of sphere.

Vector Form

If the position vectors of the extremities of a diameter of a sphere are a and b, then its equation is

$|\vec{r}-\vec{a}|^2+|\vec{r}-\vec{b}|^2=|\vec{a}-\vec{b}|^2$

Proof :

Let a and b be the position vectors of the extremities A and B of a diameter of a sphere. Let r be the position vector of any point P on the sphere, then,

$AP = \vec{r}-\vec{a}$

and $BP = \vec{r}-\vec{b}$

Since, ΔAPB is a right angled triangle.

$\therefore \quad AP^2 + BP^2 = AB^2$

$\Rightarrow \quad |AP|^2 + |BP|^2 = |AB|^2$

$\Rightarrow \quad |\vec{r}-\vec{a}|+|\vec{r}-\vec{b}|^2=|\vec{a}-\vec{b}|^2$

This is the required equation of the sphere.

Cartesian Form

If (x_1, y_1, z_1) and (x_2, y_2, z_2) are the coordinates of the extremities of a diameter of a sphere, then its equation is, $(x-x_1)(x-x_2) + (y-y_1)(y-y_2) + (z-z_1)(z-z_2) = 0$

Q. **Find the equation of the sphere described on the joint of points A and B having Position vectors $2\hat{i}+6\hat{j}-7\hat{k}$ and $-2\hat{i}+4\hat{j}-3\hat{k}$, respectively, as the diameter.**

Find the centre and the radius of the sphere.

Sol. If point P with position vector $r = x\hat{i}+y\hat{j}+z\hat{k}$ is any point on the sphere, then

$\overline{AP}\cdot\overline{BP} = 0$.

$(x-2)(x+2) + (y-6)(y-4) + (z+3)(z+7) = 0$

$\Rightarrow (x^2-4) + (y^2-10y+24) + (z^2+10z+21) = 0$

$\Rightarrow \quad x^2+y^2+z^2-10y+10z+41 = 0$

The centre of this sphere is (0, 5, −5) and its radius is $\sqrt{5^2+(-5)^2-41} = \sqrt{9} = 3$

Section of a Sphere by a Plane

Consider a sphere intersected by a plane. The set of points common to both sphere and plane is called a plane section of a sphere. It can be easily seen the plane section of sphere is a circle.

Let C be the centre of the sphere and M be the foot of the perpendicular from C on the plane. Then, M is the centre of the circle and radius of circle is given by PM.

i.e. $PM = \sqrt{CP^2-CM^2}$

The centre M of the circle is the point of intersection of the plane and line CM, which passes through C and is perpendicular to given plane.

Q. **Find the centre of the circle given by $\vec{r}\cdot(\hat{i}+2\hat{j}+2\hat{k})=15$ and $|\vec{r}-(\hat{j}+2\hat{k})|=4$**

Sol. The equation of a line through the centre $\hat{j}+2\hat{k}$ and normal to the given plane is

$r=(\hat{j}+2\hat{k})+\lambda(\hat{i}+2\hat{j}+2\hat{k})$ ...(i)

This meets the plane at a point for which we must have

$(\hat{j}+2\hat{k})+\lambda(\hat{i}+2\hat{j}+2\hat{k})]\cdot(\hat{i}+2\hat{j}+2\hat{k})=15$

$\Rightarrow \quad 6+9\lambda=15$

$\Rightarrow \quad \lambda=1$

On putting $\lambda=1$ in Eq. (i), we obtain the position vectors of the centre as $\hat{i}+3j+4\hat{k}$, Hence, the coordinates of the centre of the circle are (1, 3, 4)

Q. **Find the radius of circular section in which the sphere $|\vec{r}|=5$ is cut by plane $r\cdot(\hat{i}+\hat{j}+\hat{k})=3\sqrt{3}$**

Sol. Let A be the foot of the perpendicular from the centre O to the plane $r\cdot(\hat{i}+\hat{j}+\hat{k})-3\sqrt{3}=0$

Then, $|OA|=\left|\dfrac{0\cdot(\hat{i}+j+\hat{k})-3\sqrt{3}}{|\hat{i}+\hat{j}+\hat{k}|}\right|=\dfrac{3\sqrt{3}}{\sqrt{3}}=3$

(perpendicular distance of a point from the plane)

If P is any point on the circle, then P lies on the plane as well as on the sphere.

Therefore, OP = radius of the sphere = 5

Now, $AP^2=OP^2-OA^2=5^2-3^2=16$

$\Rightarrow \quad AP=4$

